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Wang

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(54) **SUB-BITSTREAM EXTRACTION OF
MULTI-LAYER VIDEO BITSTREAMS**

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CPC **H04N 19/46** (2014.11); **H04N 19/105** (2014.11); **H04N 19/188** (2014.11)
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CPC H04N 19/46; H04N 19/105; H04N 19/188
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(56) **References Cited**

U.S. PATENT DOCUMENTS

8,542,492 B2 9/2013 Dunn, Jr.
8,594,202 B2 * 11/2013 Eleftheriadis H04N 21/2662
375/240.26

(Continued)

FOREIGN PATENT DOCUMENTS

AU 2021276676 B2 8/2024
CN 105981387 A 9/2016
(Continued)

OTHER PUBLICATIONS

SVC temporal layer switching points; Wang—2007. (Year: 2007).*
(Continued)

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(57) **ABSTRACT**
Examples of video encoding methods and apparatus and video decoding methods and apparatus are described. An example method of video processing includes performing a conversion between a video and a bitstream of the video according to a rule, wherein the bitstream includes network abstraction layer (NAL) units for multiple video layers according to a rule; wherein the rule defines that a sub-bitstream extraction process to generate an output bitstream comprising an output layer set (OLS) includes one or more operations that are selectively performed responsive to the following conditions: (1) a list of NAL unit header layer identifier values in the OLS does not includes all values of NAL unit header layer identifiers in all video coding layer (VCL) NAL units in the bitstream, and (2) the output bitstream containing a supplemental enhancement information (SEI) NAL unit that contains a scalable-nesting SEI message.

16 Claims, 9 Drawing Sheets

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Performing a conversion between a video and a bitstream of the video, wherein the bitstream includes network abstraction layer (NAL) units for multiple video layers according to a rule; wherein the rule defines a sub-bitstream extraction process by which NAL units are removed from the bitstream to generate an output bitstream, and wherein the rule specifies to remove all supplemental enhancement information (SEI) NAL units that contain a non-scalable-nested SEI message with a particular payload type, responsive to a list of NAL unit header layer identifier values in an output layer set (OLS) with a target OLS index not including all values of NAL unit header layer identifiers in all video coding layer (VCL) NAL units in the bitstream that is input to the sub-bitstream extraction process.

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(56) References Cited

U.S. PATENT DOCUMENTS

9,154,785 B2 10/2015 Wang
 9,270,989 B2 * 2/2016 Hannuksela H04N 19/70
 9,521,393 B2 12/2016 Wang
 9,584,804 B2 2/2017 Wang
 9,712,833 B2 7/2017 Wang
 9,894,370 B2 2/2018 Wang
 10,205,965 B2 * 2/2019 Hannuksela H04N 19/127
 10,432,951 B2 10/2019 Wang
 11,356,681 B2 * 6/2022 Choi H04N 19/105
 11,825,102 B2 * 11/2023 Wang H04N 21/4345
 11,856,215 B2 * 12/2023 Wang H04N 19/37
 11,962,790 B2 * 4/2024 Wang H04N 21/4347
 12,108,068 B2 10/2024 Wang
 12,192,493 B2 1/2025 Wang
 2002/0136298 A1 9/2002 Anantharamu
 2010/0232508 A1 9/2010 Kang
 2013/0191550 A1 7/2013 Hannuksela
 2014/0003489 A1 1/2014 Hannuksela
 2014/0086341 A1 3/2014 Wang
 2014/0098894 A1 4/2014 Wang
 2014/0098895 A1 4/2014 Wang
 2014/0098896 A1 4/2014 Wang
 2014/0192149 A1 7/2014 Wang
 2014/0301476 A1 10/2014 Deshpande
 2015/0103888 A1 4/2015 Chen
 2015/0103921 A1 4/2015 Hannuksela
 2015/0103927 A1 4/2015 Hannuksela
 2015/0131744 A1 5/2015 Samuelsson
 2015/0172679 A1 6/2015 Choi
 2015/0189322 A1 7/2015 He
 2015/0271498 A1 9/2015 Wang
 2015/0271528 A1 9/2015 Wang
 2015/0271529 A1 9/2015 Wang
 2015/0358640 A1 12/2015 Hendry
 2015/0373346 A1 12/2015 Wang
 2016/0255373 A1 9/2016 Deshpande
 2016/0309159 A1 10/2016 Deshpande
 2016/0366435 A1 12/2016 Chien et al.
 2017/0019673 A1 1/2017 Tsukuba
 2017/0214923 A1 7/2017 He
 2018/0115787 A1 4/2018 Koo
 2018/0255296 A1 9/2018 Wang
 2018/0278964 A1 9/2018 Wang
 2019/0014337 A1 1/2019 Skupin
 2019/0174144 A1 6/2019 Hannuksela
 2021/0203965 A1 7/2021 Choi et al.
 2021/0329303 A1 10/2021 Deshpande
 2022/0217375 A1 7/2022 Wang
 2022/0217387 A1 7/2022 Wang
 2022/0217391 A1 7/2022 Wang
 2022/0217411 A1 7/2022 Wang
 2022/0217412 A1 7/2022 Wang
 2023/0063674 A1 3/2023 Wang
 2023/0065580 A1 3/2023 Sanchez De La Fuente
 2023/0079188 A1 3/2023 Wang
 2023/0095551 A1 3/2023 Wang
 2023/0106638 A1 4/2023 Wang
 2023/0232030 A1 7/2023 Wang

FOREIGN PATENT DOCUMENTS

EP 3123724 B1 2/2020
 IN 548033 8/2024
 JP 2017510100 A 4/2017
 JP 2017510198 A 4/2017
 JP 2017522779 A 8/2017
 JP 2017525234 A5 7/2018
 JP 2022537576 A 8/2022
 JP 7529807 B2 8/2024
 JP 7549045 B2 9/2024
 KR 20170015916 A 2/2017
 KR 20170022994 A 3/2017
 KR 102734875 B1 11/2024
 MX 418490 11/2024
 RU 2612577 C2 3/2017
 WO 2014106685 A1 7/2014
 WO 2015056182 A2 4/2015
 WO 2020103931 A1 5/2020
 WO 2020254720 A1 12/2020
 WO 2021022273 A1 2/2021
 WO 2021237126 A1 11/2021

OTHER PUBLICATIONS

Showcase of temporal level nesting flag; Eleftheriadis—2007. (Year: 2007). *

Showcase of temporal level switching point SEI message; Wang—2007. (Year: 2007). *

Document: JVET-U1004-v1, Chris Rosewarne, “Errata report items for VVC, VSEI, HEVC, AVC, Video CICC, and CP Usage TR,” Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29, 21st Meeting, by teleconference, Jan. 6-15, 2021, 35 pages.

Office Action from Japanese Patent Application No. 2022-571347 mailed May 2, 2024 (23 pages).

Korean Notice of Allowance from Korean Patent Application No. 10-2022-7043879 dated Aug. 23, 2024, 5 pages.

Singapore Office Action from Singapore Patent Application No. 11202260459R dated Jul. 23, 2024, 8 pages.

Russian Office Action from Russian Patent Application No. 2022130182/07 dated Sep. 6, 2024, 11 pages.

Wenger, S., et al. “RTP payload format for H. 264/SVC scalable video coding.” Journal of Zhejiang University—Science A 7.5 (2006), Feb. 25, 2006, pp. 657-667.

Schwarz, H., et al., “Overview of the scalable video coding extension of the H. 264/AVC standard.” IEEE Transactions on circuits and systems for video technology vol. 17 No. 9, Sep. 2007, 18 pages.

Document: JVET-R2001-vA, Bross, B., et al., “Versatile Video Coding (Draft 9),” Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 18th Meeting: by teleconference, Apr. 15-24, 2020, 524 pages. Relevant pp. 462-464.
 Document: JVET-R2001-vA, Bross, B., et al., “Versatile Video Coding (Draft 9),” Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 18th Meeting: by teleconference, Apr. 15-24, 2020, 488 pages. H.266 Draft 9, Apr. 2020, Bross et al. (Year: 2020) Relevant pp. 462 143.

Document: JVET-R0090-v2, Bross, B., et al., “AHG9: On Video Usability Information,” Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 18th Meeting: by teleconference, Apr. 15-24, 2020, 3 pages.

“Rickard Sjöberg et al: “HEVC High-Level Syntax” In: “High Efficiency Video Coding (HEVC): Algorithms and Architectures”, Aug. 23, 2014 (Aug. 23, 2014), Springer, XP055580157, ISBN: 978-3-319-06895-4 pp. 13-48, DOI: 10.1007/978-3-319-06895-4_2, * abstract * sections 2.1-2.2, with subsections *.”

Document: JVET-P2001-vE, Bross, B., et al., “Versatile Video Coding (Draft 7),” Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 16th Meeting: Geneva, CH, Oct. 1-11, 2019, 18 pages.

Document: JVET-S0097, Drugeon, V., “Specification for JVET-S0097,” Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 18th Meeting: by teleconference, Apr. 15-24, 2020, 14 pages.

(56)

References Cited**OTHER PUBLICATIONS**

Document: JVET-S0158-v1, Wang, Y.K., et al., "AHG9/AHG8: On the general sub-bitstream extraction process," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 19th Meeting: by teleconference, Jun. 22-Jul. 1, 2020, 4 pages. JVT-X032_ SVC temporal layer switching points; Wang—2007. (Year: 2007).

JVT-V087_ Showcase of temporal level nesting flag; Eleftheriadis—2007. (Year: 2007).

JVT-Y046_ Showcase of temporal level switching point SEI message; Wang—2007. (Year: 2007).

Foreign Communication From A Related Counterpart Communication, PCT Application No. PCT/US2021/033648 dated Aug. 24, 2021, 13 pages.

Foreign Communication From A Related Counterpart Communication, PCT Application No. PCT/US2021/033707 dated Sep. 14, 2021, 16 pages.

Foreign Communication From A Related Counterpart Communication, PCT Application No. PCT/US2021/033712 dated Aug. 16, 2021, 13 pages.

Foreign Communication From A Related Counterpart Communication, PCT Application No. PCT/US2021/033717 dated Aug. 24, 2021, 15 pages.

Foreign Communication From A Related Counterpart Communication, PCT Application No. PCT/US2021/036471 dated Sep. 1, 2021, 28 pages.

Foreign Communication From A Related Counterpart Communication, PCT Application No. PCT/US/2021/036473 dated Sep. 16, 2021, 35 pages.

Foreign Communication From A Related Counterpart Communication, Indian Application No. 202247071766 dated Feb. 6, 2023, 5 pages.

Foreign Communication From A Related Counterpart Application, Indian Application No. 202247071727 dated Feb. 3, 2023, 6 pages.

Non-Final Office Action dated Mar. 23, 2023, 17 pages, U.S. Appl. No. 17/992,176, filed Nov. 22, 2022.

Notice of Allowance dated Apr. 14, 2023, 15 pages, U.S. Appl. No. 18/077,803, filed Dec. 8, 2022.

Foreign Communication From A Counterpart Application, European Application No. 21808808.6, Extended European Search Report dated Jun. 12, 2023, 8 pages.

Non-Final Office Action dated Jul. 6, 2023, 16 pages, U.S. Appl. No. 17/992,181, filed Nov. 22, 2022.

Foreign Communication From A Related Counterpart Application, Indian Application No. 202247067002, Indian Office Action dated Apr. 26, 2023.

Notice of Allowance dated Aug. 11, 2023, 28 pages, U.S. Appl. No. 18/077,880, filed Dec. 8, 2022.

Non-Final Office Action dated Apr. 6, 2023, 18 pages, U.S. Appl. No. 18/077,880, filed Dec. 8, 2022.

Extended European Search Report from European Application No. 21808542.1 dated Sep. 12, 2023.

Non-Final Office Action from U.S. Appl. No. 18/527,601 dated Nov. 6, 2024, 28 pages.

Office Action for Vietnamese Patent Application No. 1-2022-08121, dated Oct. 31, 2024, 2 pages.

First Office Action for Chinese Application No. 202180037274.6, mailed on Dec. 16, 2024, 14 pages.

Chen J., et al., "Algorithm Description of Joint Exploration Test Model 7 (JEM 7)," Joint Video Exploration Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC1/SC 29/WG 11, 7th Meeting: Torino, IT, Jul. 13-21, 2017, Document: JVET-G1001, 51 Pages.

Document: JVET-R2001-vB, Bross B. et al., "Versatile Video Coding (Draft 9)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 18th Meeting: by teleconference, 1524 Apr. 2020, 530 pages.

Extended European Search Report for European Application No. 218222446, mailed Oct. 10, 2023, 9 Pages.

Chinese Office Action from Chinese Patent Application No. 202180041836.4 dated Feb. 7, 2025, 17 pages.

Brazilian Office Action from Brazilian Patent Application No. BR112022025149-1 dated May 27, 2025, 11 pages.

* cited by examiner

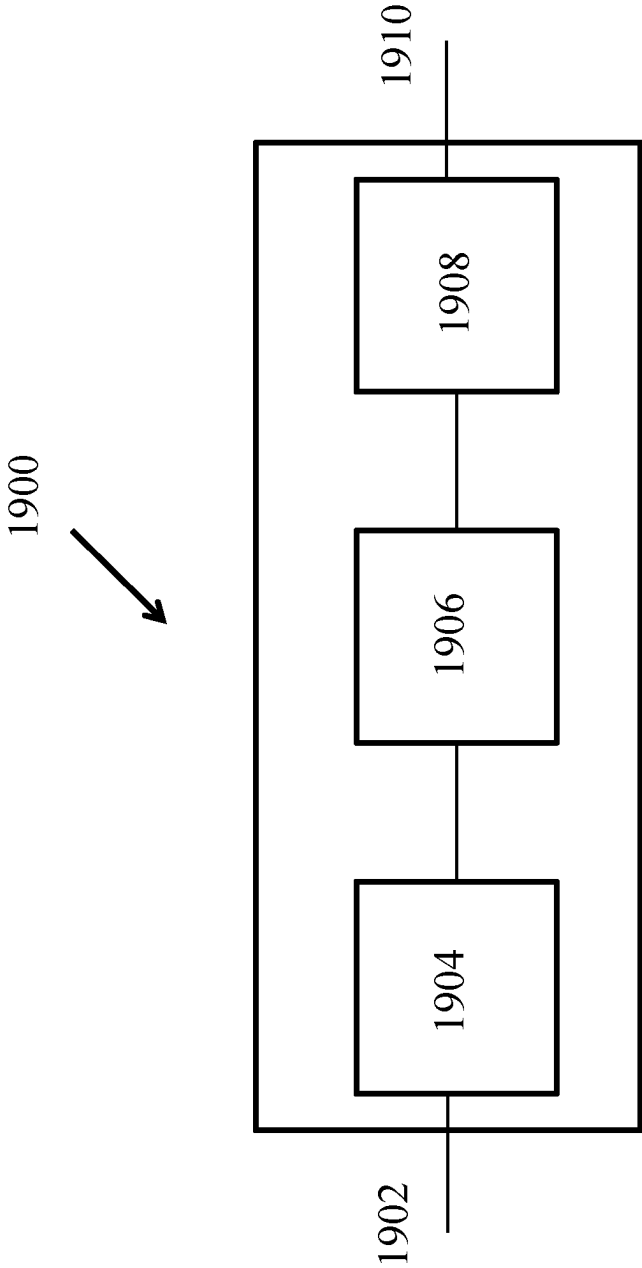


FIG. 1

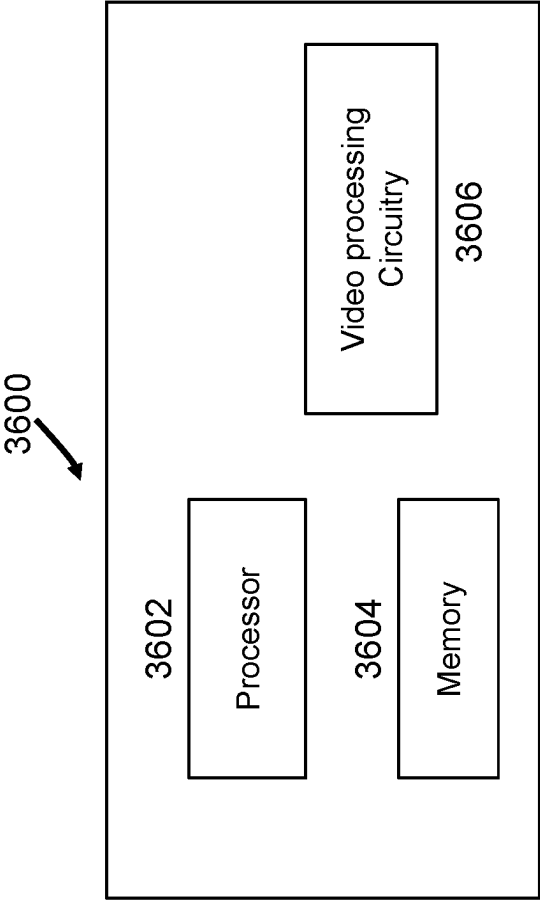
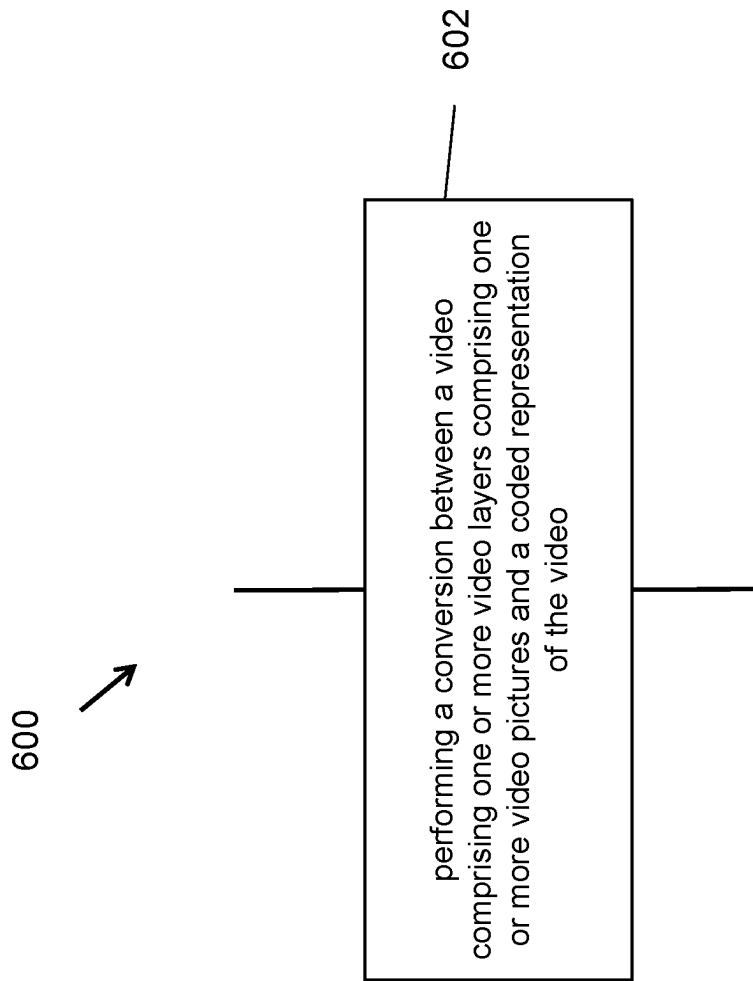


FIG. 2

**FIG. 3**

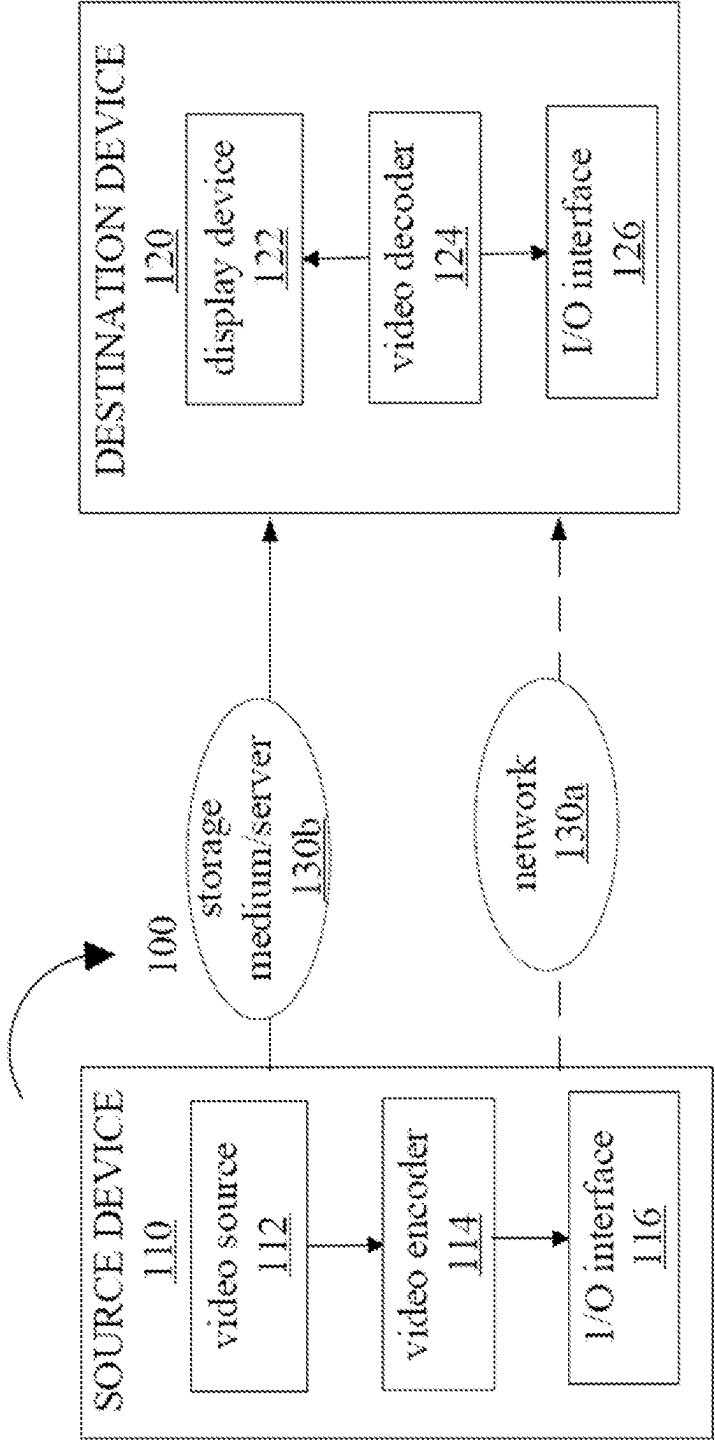


FIG. 4

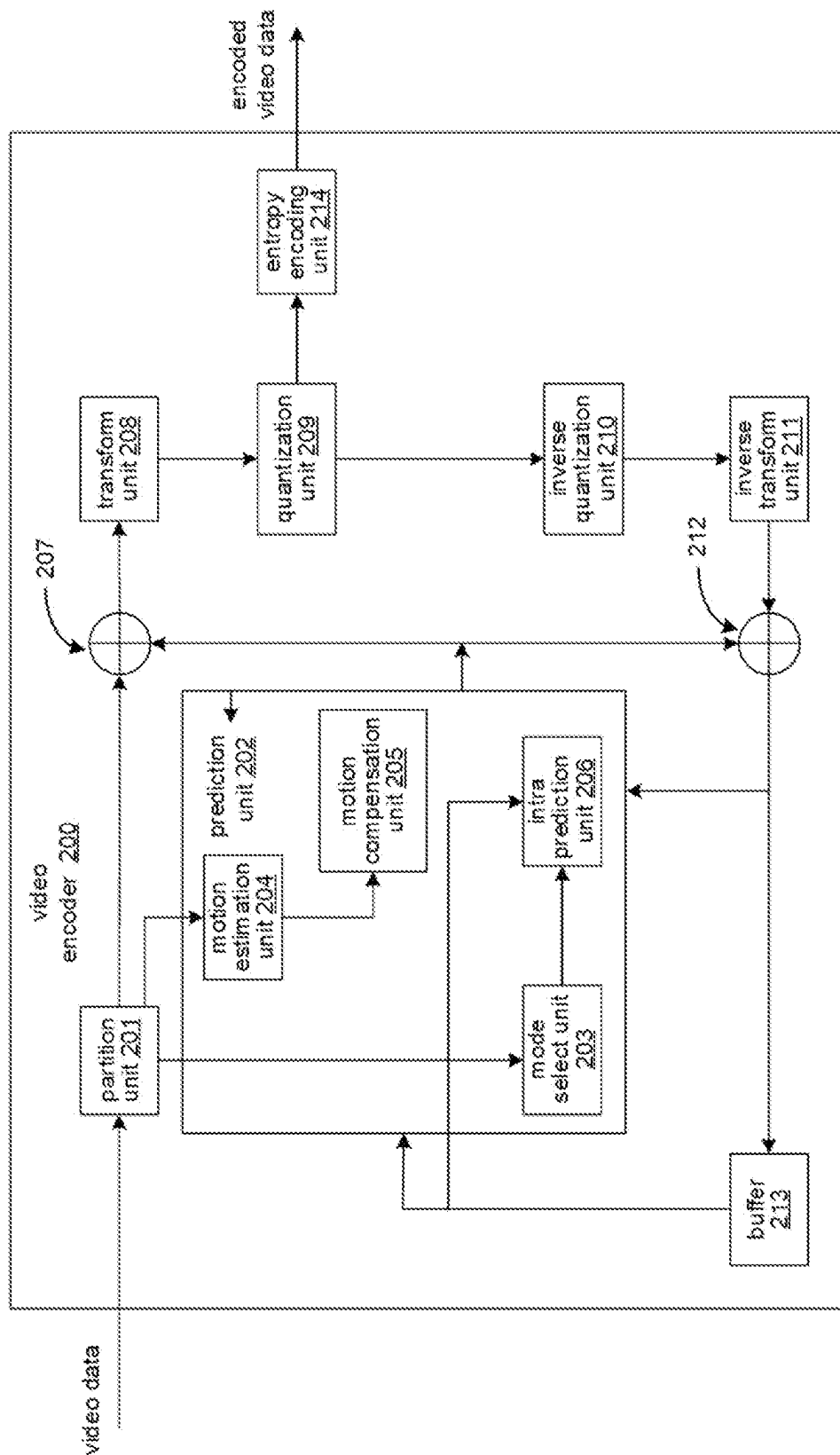


FIG. 5

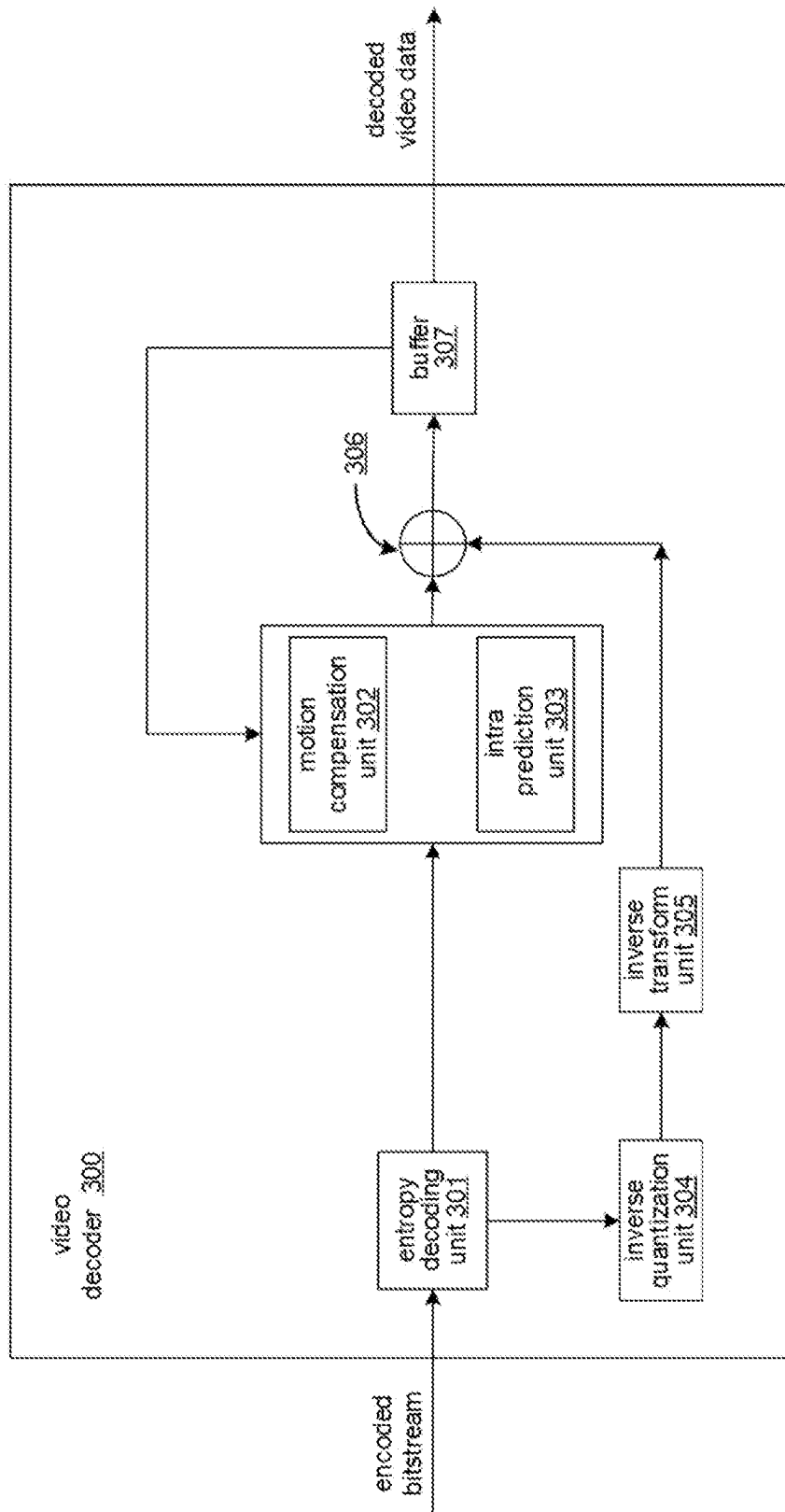


FIG. 6

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Performing a conversion between a video and a bitstream of the video, wherein the bitstream includes network abstraction layer (NAL) units for multiple video layers according to a rule; wherein the rule defines a sub-bitstream extraction process by which NAL units are removed from the bitstream to generate an output bitstream, and

wherein the rule specifies to remove all supplemental enhancement information (SEI) NAL units that contain a non-scalable-nested SEI message with a particular payload type, responsive to a list of NAL unit header layer identifier values in an output layer set (OLS) with a target OLS index not including all values of NAL unit header layer identifiers in all video coding layer (VCL) NAL units in the bitstream that is input to the sub-bitstream extraction process.

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FIG. 7A

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Performing a conversion between a video and a bitstream of the video, wherein the bitstream is separable into one or more sub-bitstreams according to a rule that specifies a sub-bitstream extraction process to generate an output bitstream, and wherein the rule specifies whether or how to remove, based on a type of a video coding layer (VCL) network abstraction layer (NAL) unit and a temporal identifier of a video coding layer associated with the VCL NAL unit, all supplemental enhancement information (SEI) network abstraction layer (NAL) units that contain an SEI message that apply to a picture or a subpicture for which the VCL NAL unit is removed during the sub-bitstream extraction process.

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FIG. 7B

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Performing a conversion between a video and a bitstream of the video according to a rule,
wherein the bitstream includes network abstraction layer (NAL) units for multiple video layers according to a rule; wherein the rule defines that a sub-bitstream extraction process to generate an output bitstream comprising an output layer set (OLS) includes one or more operations that are selectively performed responsive to the following conditions: (1) a list of NAL unit header layer identifier values in the OLS does not include all values of NAL unit header layer identifiers in all video coding layer (VCL) NAL units in the bitstream, and (2) the output bitstream containing a supplemental enhancement information (SEI) NAL unit that contains a scalable-nesting SEI message.

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FIG. 8

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SUB-BITSTREAM EXTRACTION OF MULTI-LAYER VIDEO BITSTREAMS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation application of U.S. application Ser. No. 18/077,880, filed on Dec. 8, 2022, which is a continuation of International Patent Application No. PCT/US2021/036473 filed on Jun. 8, 2021, which claims the priority to and benefits of U.S. Provisional Patent Application No. 63/036,865, filed on Jun. 9, 2020. All the aforementioned patent applications are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

This patent document relates to image and video coding and decoding.

BACKGROUND

Digital video accounts for the largest bandwidth use on the internet and other digital communication networks. As the number of connected user devices capable of receiving and displaying video increases, it is expected that the bandwidth demand for digital video usage will continue to grow.

SUMMARY

The present document discloses techniques that can be used by video encoders and decoders to perform video encoding or decoding.

In one example aspect, a video processing method is disclosed. The method includes performing a conversion between a video and a bitstream of the video, wherein the bitstream includes network abstraction layer (NAL) units for multiple video layers according to a rule; wherein the rule defines a sub-bitstream extraction process by which NAL units are removed from the bitstream to generate an output bitstream, and wherein the rule specifies to remove all supplemental enhancement information (SEI) NAL units that contain a non-scalable-nested SEI message with a particular payload type, responsive to a list of NAL unit header layer identifier values in an output layer set (OLS) with a target OLS index not including all values of NAL unit header layer identifiers in all video coding layer (VCL) NAL units in the bitstream that is input to the sub-bitstream extraction process.

In another example aspect, another video processing method is disclosed. The method includes performing a conversion between a video and a bitstream of the video, wherein the bitstream is separable into one or more sub-bitstreams according to a rule that specifies a sub-bitstream extraction process to generate an output bitstream, and wherein the rule specifies whether or how to remove, based on a type of a video coding layer (VCL) network abstraction layer (NAL) unit and a temporal identifier of a video coding layer associated with the VCL NAL unit, all supplemental enhancement information (SEI) network abstraction layer (NAL) units that contain an SEI message that apply to a picture or a subpicture for which the VCL NAL unit is removed during the sub-bitstream extraction process.

In another example aspect, another video processing method is disclosed. The method includes performing a conversion between a video and a bitstream of the video

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according to a rule, wherein the bitstream includes network abstraction layer (NAL) units for multiple video layers according to a rule; wherein the rule defines that a sub-bitstream extraction process to generate an output bitstream comprising an output layer set (OLS) includes one or more operations that are selectively performed responsive to the following conditions: (1) a list of NAL unit header layer identifier values in the OLS does not include all values of NAL unit header layer identifiers in all video coding layer (VCL) NAL units in the bitstream, and (2) the output bitstream containing a supplemental enhancement information (SEI) NAL unit that contains a scalable-nesting SEI message.

In yet another example aspect, a video encoder apparatus is disclosed. The video encoder comprises a processor configured to implement above-described methods.

In yet another example aspect, a video decoder apparatus is disclosed. The video decoder comprises a processor configured to implement above-described methods.

In yet another example aspect, a computer readable medium having code stored thereon is disclosed. The code embodies one of the methods described herein in the form of processor-executable code.

These, and other, features are described throughout the present document.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a block diagram that illustrates a video coding system in accordance with some implementations of the disclosed technology.

FIG. 2 is a block diagram of an example hardware platform used for video processing.

FIG. 3 is a flowchart for an example method of video processing.

FIG. 4 is a block diagram that illustrates an example video coding system.

FIG. 5 is a block diagram that illustrates an encoder in accordance with some implementations of the disclosed technology.

FIG. 6 is a block diagram that illustrates a decoder in accordance with some implementations of the disclosed technology.

FIGS. 7A and 7B are flowcharts for example methods of video processing based on some implementations of the disclosed technology.

FIG. 8 is a flowchart for an example method of video processing based on some implementations of the disclosed technology.

DETAILED DESCRIPTION

Section headings are used in the present document for ease of understanding and do not limit the applicability of techniques and embodiments disclosed in each section only to that section. Furthermore, H.266 terminology is used in some description only for ease of understanding and not for limiting scope of the disclosed techniques. As such, the techniques described herein are applicable to other video codec protocols and designs also.

1. Introduction

This document is related to video coding technologies. Specifically, it is about some improvements on the general sub-bitstream extraction process, signalling of picture-level Hypothetical Reference Decoder (HRD) parameters, and

containing of Supplemental Enhancement Information (SEI) messages in SEI Network Abstraction Layer (NAL) units. The ideas may be applied individually or in various combination, to any video coding standard or non-standard video codec that supports multi-layer video coding, e.g., the being-developed Versatile Video Coding (VVC).

2. Abbreviations

APS Adaptation Parameter Set
 AU Access Unit
 AUD Access Unit Delimiter
 AVC Advanced Video Coding
 CLVS Coded Layer Video Sequence
 CPB Coded Picture Buffer
 CRA Clean Random Access
 CTU Coding Tree Unit
 CVS Coded Video Sequence
 DCI Decoding Capability Information
 DPB Decoded Picture Buffer
 EOB End Of Bitstream
 EOS End Of Sequence
 GDR Gradual Decoding Refresh
 HEVC High Efficiency Video Coding
 HRD Hypothetical Reference Decoder
 IDR Instantaneous Decoding Refresh
 ILP Inter-Layer Prediction
 ILRP Inter-Layer Reference Picture
 JEM Joint Exploration Model
 LTRP Long-Term Reference Picture
 MCTS Motion-Constrained Tile Sets
 NAL Network Abstraction Layer
 OLS Output Layer Set
 PH Picture Header
 PPS Picture Parameter Set
 PTL Profile, Tier and Level
 PU Picture Unit
 RAP Random Access Point
 RBSP Raw Byte Sequence Payload
 SEI Supplemental Enhancement Information
 SPS Sequence Parameter Set
 STRP Short-Term Reference Picture
 SVC Scalable Video Coding
 VCL Video Coding Layer
 VPS Video Parameter Set
 VTM VVC Test Model
 VUI Video Usability Information
 VVC Versatile Video Coding

3. Initial discussion

Video coding standards have evolved primarily through the development of the well-known International Telecommunication Union-Telecommunication Standardization Sector (ITU-T) and International Organization for Standardization (ISO)/International Electrotechnical Commission (IEC) standards. The ITU-T produced H.261 and H.263, ISO/IEC produced Moving Picture Experts Group (MPEG)-1 and MPEG-4 Visual, and the two organizations jointly produced the H.262/MPEG-2 Video and H.264/MPEG-4 Advanced Video Coding (AVC) and H.265/High Efficiency Video Coding (HEVC) standards. Since H.262, the video coding standards are based on the hybrid video coding structure wherein temporal prediction plus transform coding are utilized. To explore the future video coding technologies beyond HEVC, the Joint Video Exploration Team (JVET) was founded by Video Coding Experts Group (VCEG) and

MPEG jointly in 2015. Since then, many new methods have been adopted by JVET and put into the reference software named Joint Exploration Model (JEM). The JVET meeting is concurrently held once every quarter, and the new coding standard is targeting at 50% bitrate reduction as compared to HEVC. The new video coding standard was officially named as Versatile Video Coding (VVC) in the April 2018 JVET meeting, and the first version of VVC test model (VTM) was released at that time. As there are continuous effort contributing to VVC standardization, new coding techniques are being adopted to the VVC standard in every JVET meeting. The VVC working draft and test model VTM are then updated after every meeting. The VVC project is now aiming for technical completion (FDIS) at the July 2020 meeting.

3.1. Picture Resolution Change Within a Sequence

In AVC and HEVC, the spatial resolution of pictures cannot change unless a new sequence using a new SPS starts, with an Intra Random Access Point (TRAP) picture. VVC enables picture resolution change within a sequence at a position without encoding an IRAP picture, which is always intra-coded. This feature is sometimes referred to as reference picture resampling (RPR), as the feature needs resampling of a reference picture used for inter prediction when that reference picture has a different resolution than the current picture being decoded.

The scaling ratio is restricted to be larger than or equal to 1/2 (2 times downsampling from the reference picture to the current picture), and less than or equal to 8 (8 times upsampling). Three sets of resampling filters with different frequency cutoffs are specified to handle various scaling ratios between a reference picture and the current picture. The three sets of resampling filters are applied respectively for the scaling ratio ranging from 1/2 to 1/1.75, from 1/1.75 to 1/1.25, and from 1/1.25 to 8. Each set of resampling filters has 16 phases for luma and 32 phases for chroma which is same to the case of motion compensation interpolation filters. Actually the normal MC interpolation process is a special case of the resampling process with scaling ratio ranging from 1/1.25 to 8. The horizontal and vertical scaling ratios are derived based on picture width and height, and the left, right, top and bottom scaling offsets specified for the reference picture and the current picture.

Other aspects of the VVC design for support of this feature that are different from HEVC include: i) The picture resolution and the corresponding conformance window are signaled in the PPS instead of in the SPS, while in the SPS the maximum picture resolution is signaled. ii) For a single-layer bitstream, each picture store (a slot in the Decoded Picture Buffer (DPB) for storage of one decoded picture) occupies the buffer size as required for storing a decoded picture having the maximum picture resolution.

3.2. Scalable Video Coding (SVC) in General and in VVC

Scalable video coding (SVC, sometimes also just referred to as scalability in video coding) refers to video coding in which a base layer (BL), sometimes referred to as a reference layer (RL), and one or more scalable enhancement layers (ELs) are used. In SVC, the base layer can carry video data with a base level of quality. The one or more enhancement layers can carry additional video data to support, for example, higher spatial, temporal, and/or signal-to-noise (SNR) levels. Enhancement layers may be defined relative

to a previously encoded layer. For example, a bottom layer may serve as a BL, while a top layer may serve as an EL. Middle layers may serve as either ELs or RLs, or both. For example, a middle layer (e.g., a layer that is neither the lowest layer nor the highest layer) may be an EL for the layers below the middle layer, such as the base layer or any intervening enhancement layers, and at the same time serve as a RL for one or more enhancement layers above the middle layer. Similarly, in the Multiview or 3D extension of the HEVC standard, there may be multiple views, and information of one view may be utilized to code (e.g., encode or decode) the information of another view (e.g., motion estimation, motion vector prediction and/or other redundancies).

In SVC, the parameters used by the encoder or the decoder are grouped into parameter sets based on the coding level (e.g., video-level, sequence-level, picture-level, slice level, etc.) in which they may be utilized. For example, parameters that may be utilized by one or more coded video sequences of different layers in the bitstream may be included in a video parameter set (VPS), and parameters that are utilized by one or more pictures in a coded video sequence may be included in a sequence parameter set (SPS). Similarly, parameters that are utilized by one or more slices in a picture may be included in a picture parameter set (PPS), and other parameters that are specific to a single slice may be included in a slice header. Similarly, the indication of which parameter set(s) a particular layer is using at a given time may be provided at various coding levels.

Thanks to the support of reference picture resampling (RPR) in VVC, support of a bitstream containing multiple layers, e.g., two layers with Standard Definition (SD) and High Definition (HD) resolutions in VVC can be designed without the need any additional signal-processing-level coding tool, as upsampling needed for spatial scalability support can just use the RPR upsampling filter. Nevertheless, high-level syntax changes (compared to not supporting scalability) are needed for scalability support. Scalability support is specified in VVC version 1. Different from the scalability supports in any earlier video coding standards, including in extensions of AVC and HEVC, the design of VVC scalability has been made friendly to single-layer decoder designs as much as possible. The decoding capability for multi-layer bitstreams are specified in a manner as if there were only a single layer in the bitstream. E.g., the decoding capability, such as DPB size, is specified in a manner that is independent of the number of layers in the bitstream to be decoded. Basically, a decoder designed for single-layer bitstreams does not need much change to be able to decode multi-layer bitstreams. Compared to the designs of multi-layer extensions of AVC and HEVC, the High Level Syntax (HLS) aspects have been significantly simplified at the sacrifice of some flexibilities. For example, an IRAP Access Unit (AU) is required to contain a picture for each of the layers present in the Coded Video Sequence (CVS).

3.3. Parameter Sets

AVC, HEVC, and VVC specify parameter sets. The types of parameter sets include SPS, PPS, Adaptation Parameter Set (APS), and VPS. SPS and PPS are supported in all of AVC, HEVC, and VVC. VPS was introduced since HEVC and is included in both HEVC and VVC. APS was not included in AVC or HEVC but is included in the latest VVC draft text.

SPS was designed to carry sequence-level header information, and PPS was designed to carry infrequently chang-

ing picture-level header information. With SPS and PPS, infrequently changing information need not to be repeated for each sequence or picture, hence redundant signalling of this information can be avoided. Furthermore, the use of SPS and PPS enables out-of-band transmission of the important header information, thus not only avoiding the need for redundant transmissions but also improving error resilience.

VPS was introduced for carrying sequence-level header information that is common for all layers in multi-layer bitstreams.

APS was introduced for carrying such picture-level or slice-level information that needs quite some bits to code, can be shared by multiple pictures, and in a sequence there can be quite many different variations.

3.4. General Sub-Bitstream Extraction Process

Clause C.6 of the latest VVC text specifies a general sub-bitstream extraction process, as follows:

C.6 Sub-bitstream extraction process

Inputs to this process are a bitstream `inBitstream`, a target OLS index `targetOlsIdx`, and a target highest TemporalId value `tIdTarget`.

Output of this process is a sub-bitstream `outBitstream`.

It is a requirement of bitstream conformance for the input bitstream that any output sub-bitstream that satisfies all of the following conditions shall be a conforming bitstream:

The output sub-bitstream is the output of the process specified in this clause with the bitstream, `targetOlsIdx` equal to an index to the list of OLSs specified by the VPS, and `tIdTarget` equal to any value in the range of 0 to 6, inclusive, as inputs.

The output sub-bitstream contains at least one VCL NAL unit with `nuh_layer_id` equal to each of the `nuh_layer_id` values in `LayerIdInOls[targetOlsIdx]`.

The output sub-bitstream contains at least one VCL NAL unit with `TemporalId` equal to `tIdTarget`.

NOTE—A conforming bitstream contains one or more coded slice NAL units with `TemporalId` equal to 0, but does not have to contain coded slice NAL units with `nuh_layer_id` equal to 0.

The output sub-bitstream `OutBitstream` is derived as follows:

1. The bitstream `outBitstream` is set to be identical to the bitstream `inBitstream`.
2. Remove from `outBitstream` all NAL units with `TemporalId` greater than `tIdTarget`.
3. Remove from `outBitstream` all NAL units with `nal_unit_type` not equal to any of `VPS_NUT`, `DCI_NUT`, and `EOB_NUT` and with `nuh_layer_id` not included in the list `LayerIdInOls[targetOlsIdx]`.
4. Remove from `outBitstream` all VCL NAL units for which all of the following conditions are true, and their associated non-VCL NAL units with `nal_unit_type` equal to `PH_NUT`, `FD_NUT`, `SUFFIX_SEI_NUT`, and `PREFIX_SEI_NUT` with `PayloadType` not equal to 0, 1, or 130:

`nal_unit_type` is equal to `TRAIL_NUT`, `STSA_NUT`, `RADL_NUT`, or `RASL_NUT`, or `nal_unit_type` is equal to `GDR_NUT` and the associated `ph_recovery_poc_cnt` is not equal to 0.

`nuh_layer_id` is equal to `LayerIdInOls[targetOlsIdx][j]` for a value of `j` in the range of 0 to `NumLayersInOls[targetOlsIdx]-1` inclusive.

`TemporalId` is greater than or equal to `NumSubLayersInLayerInOLS[targetOlsIdx][GeneralLayerIdx[nuh_layer_id]]`.

5. Remove from outBitstream all SEI NAL units that contain a scalable nesting SEI message that has sn_ols_flag equal to 1 and there is no value of i in the range of 0 to sn_num_olss_minus 1, inclusive, such that NestingOlsIdx[i] is equal to targetOlsIdx. 5
6. When LayerIdInOls[targetOlsIdx] does not include all values of nuh_layer_id in all NAL units in the bitstream, the following applies:
 - a. Remove from outBitstream all SEI NAL units that contain a non-scalable-nested SEI message with payloadType equal to 0 (BP) or 130 (DUI). 10
 - b. When general_same_pic_timing_in_all_ols_flag is equal to 0, remove from outBitstream all SEI NAL units that contain a non-scalable-nested SEI message with payloadType equal to 1 (PT). 15
 - c. When outBitstream contains SEI NAL units that contain a scalable nesting SEI message with sn_ols_flag equal to 1 and are applicable to outBitstream (NestingOlsIdx[i] is equal to targetOlsIdx), the following applies: 20
 - 1) If general_same_pic_timing_in_all_ols_flag is equal to 0, extract appropriate non-scalable-nested SEI message with payloadType equal to 0 (BP), 1 (PT), or 130 (DUI) from the scalable nesting SEI message and include those SEI messages in outBitstream. 25
 - Otherwise (general_same_pic_timing_in_all_ols_flag is equal to 1), extract appropriate non-scalable-nested SEI message with payloadType equal to 0 (BP) or 130 (DUI) from the scalable nesting SEI message and include those SEI messages in outBitstream. 30

4. Technical Problems Solved by Disclosed Technical Solutions 35

The existing designs of the general sub-bitstream extraction process and related other parts in the latest VVC text (in JVET-R2001-vA/v10) have the following problems:

- 1) In the conditions under which an output sub-bitstream is required to be a conforming bitstream, the value of tIdTarget is said to be in the range of 0 to 6, inclusive. However, in many bitstreams, the highest Temporal value is less than 6, and that value is specified by the syntax element vps_max_sublayers_minus1. 40
- 2) An access unit delimiter (AUD) NAL unit, when present, can have any nuh_layer_id value. However, step 3 of the sub-bitstream extraction process would remove the AUD NAL units for which the nuh_layer_id values are not included in the list LayerIdInOls[targetOlsIdx]. 45
- 3) Some SEI NAL units contain a scalable nesting SEI message with sn_ols_flag equal to 0 while the applicable layers as indicated in the scalable nesting SEI message does not include any layer in the target OLS, i.e., none of the applicable layers' nuh_layer_id values is not included in the list LayerIdInOls1 targetOlsIdx 1. These SEI NAL units should also be removed. 50
- 4) The condition of step 6, i.e., "When LayerIdInOls[targetOlsIdx] does not include all values of nuh_layer_id in all NAL units in the bitstream" has the following two issues. 60
 - a. The condition does not work for cases when Decoding Capability Information (DCI), VPS, AUD, or EOB NAL units are present and have nuh_layer_id not equal to any of the nuh_layer_id values of the VCL NAL units. 65

- b. The phrase "the bitstream" is not clear, as there are two bitstreams involved in the context, inBitstream and outBitstream.
- 5) Step 6.c would extract scalable-nested SEI messages, to generate non-scalable-nested SEI messages, from scalable nesting SEI messages with both sn_ols_flag equal to 1 and sn_subpic_flag equal to 1, while such scalable-nested SEI messages only apply to specific subpictures and thus should not be extracted.
- 6) In step 6.c, when multiple scalable-nested SEI messages are extracted from one SEI NAL unit seiNalUnitA to be non-scalable-nested SEI messages, they should still be included in one SEI NAL unit seiNalUnitB, and the SEI NAL unit seiNalUnitB should be included in the same Picture Unit (PU) that contained the SEI NAL unit seiNalUnitA. However, this is not specified.
- 7) Step 6.c should remove, from outBitstream, all SEI NAL units from which some SEI messages have been extracted and included as non-scalable-nested SEI messages. However, this is not specified.
- 8) There lacks a constraint such that when an SEI NAL unit contain an SEI message with payloadType equal to 0, 1, or 130, the SEI NAL unit shall not contain an SEI message with payloadType not equal to 0 (BP), 1 (PT), 130 (DUI), or 133 (scalable nesting). This causes the removal of SEI messages in step 4 involves more than just removal of SEI NAL units.
- 9) The flag general_same_pic_timing_in_all_ols_flag only specifies whether non-scalable-nested PT SEI messages apply to all OLSs. However, information carried in the DUI SEI messages are for similar purposes as in the PT SEI messages.
- 10) When removing from the output bitstream the VCL NAL units for which both of the following two conditions are true: a) nal_unit_type is equal to TRAIL_NUT, STSA_NUT, RADL_NUT, or RASL_NUT, or nal_unit_type is equal to GDR_NUT and the associated ph_recovery_poc_cnt is greater than 0, and b) Temporal is greater than or equal to NumSubLayersInLayerInOls[targetOlsIdx][GeneralLayerIdx[nuh_layer_id]], the step also removes the associated SEI NAL units containing SEI messages other than the BP, PT, or DUI SEI messages. However, some of those SEI messages removed may apply to OLSs or layers that contain pictures remaining in the output bitstream.
- 11) The subpicture level information SEI messages, when present, apply to OLSs, like the other HRD-related SEI messages, i.e., BP, PT, DUI SEI messages. However, when LayerIdInOls[targetOlsIdx] does not include all values of nuh_layer_id in all NAL units in the bitstream, SEI NAL units that contain a non-scalable-nested SEI message with payloadType equal to 203 (i.e., subpicture level information SEI message) are not addressed in the extraction process.
- 12) The last step that makes scalable-nested SEI messages to be non-scalable-nested SEI messages has the following issues:
 - a. SEI messages in the case with sn_ols_flag equal to 0 and sn_subpic_flag equal to 0 are not covered.
 - b. Where the resulting non-scalable-nested SEI messages should be placed (in which SEI NAL unit, where the SEI NAL unit should be) in the output bitstream is unspecified.

5. A Listing of Technical Solutions and Embodiments

To solve the above problems, and others, methods as summarized below are disclosed. The items should be

considered as examples to explain the general concepts and should not be interpreted in a narrow way. Furthermore, these items can be applied individually or combined in any manner.

- 1) To solve problem 1, the conditions under which an output sub-bitstream is required to be a conforming bitstream is specified such that the value of `tIdTarget` is specified be in the range of 0 to `vps_max_sublayers_minus1`, inclusive.
 - a. Alternatively, the conditions under which an output sub-bitstream is required to be a conforming bitstream is specified such that the value of `tIdTarget` is specified be in the range of 0 to `vps_max_sublayers_minus1`, inclusive, when there is more than one layer in the input bitstream, and specified be in the range of 0 to `sps_max_sublayers_minus1`, inclusive, when there is only one layer in the input bitstream.
- 2) To solve problem 2, the general sub-bitstream extraction process is specified such that AUD NAL units are treated in the same manner as NAL units with `nal_unit_type` equal to `VPS_NUT`, `DCI_NUT`, or `EOB_NUT`. In other words, no AUD NAL unit is removed from the output bitstream `outBitstream` according to the `nuh_layer_id` value.
- 3) To solve problem 3, the general sub-bitstream extraction process is specified such that it would remove, the output bitstream `outBitstream`, SEI NAL units that contain a scalable nesting SEI message with `sn_ols_flag` equal to 0 while the applicable layers as indicated in the scalable nesting SEI message does not included any layer in the target OLS.
 - a. In one example, it is specified to remove from `outBitstream` all SEI NAL units that contain a scalable nesting SEI message that has `sn_ols_flag` equal to 0 and there is no value in the list `nestingLayerId[i]` for `i` in the range of 0 to `nestingNumLayers - 1`, inclusive, that is in the list `LayerIdInOls[targetOlsIdx]`.
- 4) To solve problem 4, the condition “When `LayerIdInOls[targetOlsIdx]` does not include all values of `nuh_layer_id` in all NAL units in the bitstream” is changed to be “When the list `LayerIdInOls[targetOlsIdx]` does not include all values of `nuh_layer_id` in all VCL NAL units in the bitstream in `inBitstream`”.
- 5) To solve problem 5, the general sub-bitstream extraction process is specified such that it only extracts scalable-nested SEI messages from scalable nesting SEI messages with both `sn_ols_flag` equal to 1 and `sn_subpic_flag` equal to 0 to generate non-scalable-nested SEI messages.
- 6) To solve problem 6, the general sub-bitstream extraction process is specified such that, when multiple scalable-nested SEI messages are extracted from one SEI NAL unit `seiNalUnitA` to be non-scalable-nested SEI messages, they are still included in one SEI NAL unit `seiNalUnitB` in the output bitstream `outBitstream`, and the SEI NAL unit `seiNalUnitB` is included in the PU that contained the SEI NAL unit `seiNalUnitA`.
- 7) To solve problem 7, the general sub-bitstream extraction process is specified such that it removes, from the output bitstream `outBitstream`, all SEI NAL units from which some SEI messages have been extracted and included as non-scalable-nested SEI messages.
 - a. Alternatively, when the scalable-nested SEI messages in such an SEI NAL unit apply only to the

- target OLS (i.e., the `targetOlsIdx`-th OLS specified by the VPS), remove the SEI NAL unit from `outBitstream`.
- b. Alternatively, when there is no OLS, other than the target OLS in the OLSs to which the scalable-nested SEI messages in such an SEI NAL unit apply, that contains layers that are all included in the list `LayerIdInOls[targetOlsIdx]`, remove the SEI NAL unit from `outBitstream`.
 - 8) To solve problem 8, add constraint such that when an SEI NAL unit contain an SEI message with `payloadType` equal to 0, 1, or 130, the SEI NAL unit shall not contain an SEI message with `payloadType` not equal to 0 (BP), 1 (PT), 130 (DUI), or 133 (scalable nesting).
 - 9) To solve problem 9, the flag `general_same_pic_timing_in_all_ols_flag` specifies whether non-scalable-nested PT and DUI SEI messages apply to all OLSs.
 - a. Alternatively, the flag `general_same_pic_timing_in_all_ols_flag` specifies whether non-scalable-nested BP, PT, and DUI SEI messages apply to all OLSs.
 - i. In one example, the flag `general_same_pic_timing_in_all_ols_flag` is renamed to be flag `general_same_pic_level_hrd_info_in_all_ols_flag`, which specifies whether non-scalable-nested BP, PT, and DUI SEI messages apply to all OLSs.
 - b. Alternatively, a new flag, e.g., named `general_same_oi_in_all_ols_flag`, is added, to specify whether non-scalable-nested DUI SEI messages apply to all OLSs.
 - c. Alternatively, a new flag, e.g., named `general_same_bp_in_all_ols_flag`, is added, to specify whether non-scalable-nested BP SEI messages apply to all OLSs.
 - 10) To solve problem 10, in one example, in the general sub-bitstream extraction process, when removing from the output bitstream the VCL NAL units for which both of the following two conditions are true: a) `nal_unit_type` is equal to `TRAIL_NUT`, `STSA_NUT`, `RADL_NUT`, or `RASL_NUT`, or `nal_unit_type` is equal to `GDR_NUT` and the associated `ph_recovery_poc_cnt` is greater than 0, and b) `Temporand` is greater than or equal to `NumSubLayersInLayerInOls[targetOlsIdx][GeneralLayerId[nuh_layer_id]]`, instead of removing the associated SEI NAL units containing SEI messages other than the BP, PT, or DUI SEI messages, remove all all SEI NAL units that contain SEI messages that only apply to one or more of the pictures or subpictures for which all VCL NAL units are being removed.
 - a. In one example, alternatively, keep the associated SEI NAL units in the output bitstream when removing such VCL NAL units.
 - 11) To solve problem 11, in one example, in the general sub-bitstream extraction process, when `LayerIdInOls[targetOlsIdx]` does not include all values of `nuh_layer_id` in all NAL units in the bitstream, additionally remove all SEI NAL units that contain a non-scalable-nested SEI message with `payloadType` equal to 203 (i.e., subpicture level information SEI message).
 - 12) To solve problem 12, in one example, in the general sub-bitstream extraction process, when `LayerIdInOls[targetOlsIdx]` does not include all values of `nuh_layer_id` in all NAL units in the bitstream, and the `outBitstream` contains an SEI NAL unit `seiNalUnitA` that contains a scalable nesting SEI message with `sn_subpic_flag` equal to 0 that applies to the OLSs (when `sn_ols_flag` is equal to 1) or the layers (when `sn_`

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ols_flag is equal to 0) that have the same set of layers as in outBitstream, perform one or more of the following:

- a. Generate a new SEI NAL unit seiNalUnitB.
- b. Include seiNalUnitB in the PU containing seiNalUnitA.
- c. Include seiNalUnitB in the PU containing seiNalUnitA immediately after seiNalUnitA.
- d. Extract the scalable-nested SEI messages from the scalable nesting SEI message and include them directly in seiNalUnitB (as non-scalable-nested SEI messages).
- e. Remove seiNalUnitA from outBitstream.

13) In one example, in the general sub-bitstream extraction process, when

LayerIdInOls[targetOlsIdx] does not include all values of nuh_layer_id in all NAL units in the bitstream, and the outBitstream contains an SEI NAL unit seiNalUnitA that contains a scalable nesting SEI message, keep seiNalUnitA in the output bitstream.

6. Embodiments

Below are some example embodiments for some of the invention aspects summarized above in Section 5, which can be applied to the VVC specification. The changed texts are based on the latest VVC text in JVET-R2001-vA/v10. Most relevant parts that have been added or modified are highlighted in bold and Italic, and some of the deleted parts are marked with double brackets (e.g., [[a]]) denotes the deletion of the character “a”). There may be some other changes that are editorial in nature and thus not highlighted.

6.1. First Embodiment

This embodiment is for items 1, 2, 3, 3.a, 4, 5, 6, 7.b, and 8.

C.6 General General sub-bitstream extraction process

Inputs to this process are a bitstream inBitstream, a target OLS index targetOlsIdx, and a target highest TemporalId value tIdTarget.

Output of this process is a sub-bitstream outBitstream.

It is a requirement of bitstream conformance for the input bitstream that any output sub-bitstream that satisfies all of the following conditions shall be a conforming bitstream:

The output sub-bitstream is the output of the process specified in this clause with the bitstream, targetOlsIdx equal to an index to the list of OLSs specified by the VPS, and tIdTarget equal to any value in the range of 0 to *vps_max_sublayers_minus1*, inclusive, as inputs.

The output sub-bitstream contains at least one VCL NAL unit with nuh_layer_id equal to each of the nuh_layer_id values in LayerIdInOls[targetOlsIdx].

The output sub-bitstream contains at least one VCL NAL unit with TemporalId equal to tIdTarget.

NOTE—A conforming bitstream contains one or more coded slice NAL units with TemporalId equal to 0, but does not have to contain coded slice NAL units with nuh_layer_id equal to 0.

The output sub-bitstream OutBitstream is derived by applying the following ordered steps:

1. The bitstream outBitstream is set to be identical to the bitstream inBitstream.
2. Remove from outBitstream all NAL units with TemporalId greater than tIdTarget.
3. Remove from outBitstream all NAL units with nal_unit_type not equal to any of DCI_NUT, VPS_

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NUT, *AUD_NUT*, and EOB_NUT and with nuh_layer_id not included in the list LayerIdInOls[targetOlsIdx].

4. Remove from outBitstream all VCL NAL units for which all of the following conditions are true, and **also remove from outBitstream**

these VCL NAL units' associated non-VCL NAL units **that have** nal_unit_type equal to PH_NUT **or** FD_NUT, **or have nal_unit_type equal to** SUFFIX_SEI_NUT **or** PREFIX_SEI_NUT **and contain SEI messages with payloadType** not equal to 0 (BP), 1 (PT), 130 (DUI), **or 133 (scalable nesting):**

nal_unit_type is equal to TRAIL_NUT, STSA_NUT, RADL_NUT, or RASL_NUT, or nal_unit_type is equal to GDR_NUT and the associated ph_recovery_poc_cnt is not equal to 0.

[[nuh_layer_id is equal to LayerIdInOls[targetOlsIdx][j] for a value of j in the range of 0 to NumLayersInOls[targetOlsIdx]–1 inclusive.]]

TemporalId is greater than or equal to NumSubLayersInLayerInOLS[targetOlsIdx][GeneralLayerId[nuh_layer_id]].

5. Remove from outBitstream all SEI NAL units that contain a scalable nesting SEI message that has sn_ols_flag equal to 1 and there is no value of i in the range of 0 to sn_num_ols_minus1, inclusive, such that NestingOlsIdx[i] is equal to targetOlsIdx.

6. Remove from outBitstream

all SEI NAL units that contain a scalable nesting SEI message that has sn_ols_flag equal to 0 and there is no value in the list nestingLayerId[i] for i in the range of 0 to nestingNumLayers – 1, inclusive, that is in the list LayerIdInOls[targetOlsIdx].

7. When LayerIdInOls[targetOlsIdx] does not include all values of nuh_layer_id in all VCL NAL units in the bitstream **in Bitstream**, the following applies **in the order listed:**

- a. Remove from outBitstream all SEI NAL units that contain a non-scalable-nested SEI message with payloadType equal to 0 (BP) or 130 (DUI).

- b. When general_same_pic_timing_in_all_ols_flag is equal to 0, remove from outBitstream all SEI NAL units that contain a non-scalable-nested SEI message with payloadType equal to 1 (PT).

- c. When outBitstream contains SEI NAL units that contain a scalable nesting SEI message with sn_ols_flag equal to 1 **and sn_subpic_flag equal to 0** that applies to the targetOlsIdx-th OLS (i.e., there is at least one value of i in the range of 0 to sn_num_ols_minus1, inclusive, such that NestingOlsIdx[i] is equal to targetOlsIdx), the following applies **in the order listed:**

- i. For each scalable-nested BP or DUI SEI message in such an SEI NAL unit seiNalUnitA, **generate a non-scalable-nested SEI message with the same payloadType and SEI payload and include it in an SEI NAL unit in the PU containing seiNalUnitA** in outBitstream.

- ii. When general_same_pic_timing_in_all_ols_flag is equal to 0, for each scalable-nested PT SEI message in such an SEI NAL unit seiNalUnitA, **generate a non-scalable-nested SEI message with the same SEI payload and**

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include it in a SEI NAL

*unit in the PU containing seiNalUnitA in out-
Bitstream.*

- iii. *When multiple SEI messages contained in a
particular such SEI NAL
unit seiNalUnitA were made
non-scalable-nested SEI messages,
those non-scalable-nested SEI
messages are included in one SEI NAL unit.*
- iv. *When there is no OLS, other than the target OLS
in the OLSs to which the scalable-nested SEI
messages in such an SEI NAL
unit seiNalUnitA apply, that contains layers that
are all included in the list LayerIdInOls/
targetOlsIdx], remove the
SEI NAL unit seiNalUnitA from outBitstream.*

D.2.2 General SEI Payload Semantics

It is a requirement of bitstream conformance that the following restrictions apply on containing of SEI messages in SEI NAL units:

- When an SEI NAL unit contains a non-scalable-nested BP SEI message, a non-scalable-nested PT SEI message, or a non-scalable-nested DUI SEI message, the SEI NAL unit shall not contain any other SEI message with payloadType not equal to 0 (BP), 1 (PT), or 130 (DUI).
- When an SEI NAL unit contains a scalable-nested BP SEI message, a scalable-nested PT SEI message, or a scalable-nested DUI SEI message, the SEI NAL unit shall not contain any other SEI message with payloadType not equal to 0 (BP), 1 (PT), 130 (DUI) or 133 (scalable nesting).
- When an SEI NAL unit contain
an SEI message with payloadType
equal to 0, 1, or 130, the SEI NAL unit
shall not contain an SEI message with payloadType
not equal to 0 (BP), 1 (PT),
130 (DUI), or 133 (scalable nesting).*

6.2. Second Embodiment

This embodiment is for items 10 to 12, inclusive, with changes relative to the embodiment 1 text above marked with highlights.

C.6 General Sub-Bitstream Extraction Process

Inputs to this process are a bitstream inBitstream, a target OLS index targetOlsIdx, and a target highest Temporal value tldTarget.

Output of this process is a sub-bitstream outBitstream.

The OLS with OLS index targetOlsIdx is referred to as the target OLS.

It is a requirement of bitstream conformance for the input bitstream that any output sub-bitstream that satisfies all of the following conditions shall be a conforming bitstream:

- The output sub-bitstream is the output of the process specified in this clause with the bitstream, targetOlsIdx equal to an index to the list of OLSs specified by the VPS, and tldTarget equal to any value in the range of 0 to vps_max_sublayers_minus1, inclusive, as inputs.
- The output sub-bitstream contains at least one VCL NAL unit with nuh_layer_id equal to each of the nuh_layer_id values in LayerIdInOls[targetOlsIdx].
- The output sub-bitstream contains at least one VCL NAL unit with TemporalId equal to tldTarget.
- NOTE—A conforming bitstream contains one or more coded slice NAL units with TemporalId equal to 0, but does not have to contain coded slice NAL units with nuh_layer_id equal to 0.

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The output sub-bitstream OutBitstream is derived by applying the following ordered steps:

1. The bitstream outBitstream is set to be identical to the bitstream inBitstream.
2. Remove from outBitstream all NAL units with TemporalId greater than tldTarget.
3. Remove from outBitstream all NAL units [[with nal_unit_type not equal to any of DCI_NUT, VPS_NUT, AUD_NUT, and EOB_NUT and with]] *that have* nuh_layer_id not included in the list LayerIdInOls[targetOlsIdx] *and are not AUD NAL units, DCI NAL units, VPS NAL units, EOB NAL units, or SEI NAL units containing non-scalable-nested SEI messages.*

4. Remove from outBitstream all VCL NAL units for which all of the following conditions are true, [[and also remove from outBitstream these VCL NAL units']] *and their* associated non-VCL NAL units that have nal_unit_type equal to PH_NUT or FD_NUT[[, or have nal_unit_type equal to SUFFIX_SEI_NUT or PREFIX_SEI_NUT and contain SEI messages with payloadType not equal to 0 (BP), 1 (PT), 130 (DUI), or 133 (scalable nesting)]]:

nal_unit_type is equal to TRAIL_NUT, STSA_NUT, RADL_NUT, or RASL_NUT, or nal_unit_type is equal to GDR_NUT and the associated ph_recovery_poc_cnt is greater than [[not equal]] to 0.

TemporalId is greater than or equal to NumSubLayersInLayerInOls[targetOlsIdx][GeneralLayerIdx[nuh_layer_id]].

5. *Remove from outBitstream all SEI NAL units that only contain SEI messages that do not apply to the OLSs, layers, pictures, subpictures, or slices in outBitstream, as follows:*

Remove from outBitstream all SEI NAL units that contain a scalable nesting SEI message that has sn_ols_flag equal to 1 and there is no value of i in the range of 0 to sn_num_ols_minus1, inclusive, such that NestingOlsIdx[i] is equal to targetOlsIdx.

Remove from outBitstream all SEI NAL units that contain a scalable nesting SEI message that has sn_ols_flag equal to 0 and there is no value in the list nestingLayerId[i] for i in the range of 0 to nestingNumLayers-1, inclusive, that is in the list LayerIdInOls[targetOlsIdx].

Alternatively, remove from

outBitstream all SEI NAL units that contain a scalable nesting SEI

message that has sn_ols_flag

equal to 0 and there is no

value in the list Nesting LayerId equal to

a value in the list LayerIdInOls[targetOlsIdx].

Remove from outBitstream all

SEI NAL units that contain

SEI messages that only apply to one or more

of the pictures or subpictures

for which all VCL NAL units have been

removed by step 4.

6. When LayerIdInOls[targetOlsIdx] does not include all values of nuh_layer_id in all VCL NAL units in the bitstream inBitstream, the following applies in the order listed:
 - a. Remove from outBitstream all SEI NAL units that contain a non-scalable-nested SEI message with payloadType equal to 0 (BP), 130 (DUI), or 203 (SLI) .

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b. When `general_same_pic_timing_in_all_ols_flag` is equal to 0, remove from `outBitstream` all SEI NAL units that contain a non-scalable-nested SEI message with `payloadType` equal to 1 (PT).

c. *When `outBitstream` contains an SEI NAL unit `seiNalUnitA` that contains a scalable nesting SEI message with `sn_subpic_flag` equal to 0 that applies to the OLSs (when `sn_ols_flag` is equal to 1) or the layers (when `sn_ols_flag` is equal to 0) that have the same set of layers as in `outBitstream`, generate a new SEI NAL unit `seiNalUnitB`, include it in the PU containing `seiNalUnitA` immediately after `seiNalUnitA`, extract the scalable-nested SEI messages from the scalable nesting SEI message and include them directly in `seiNalUnitB` (as non-scalable-nested SEI messages), and remove `seiNalUnitA` from `outBitstream`.*

[[When `outBitstream` contains SEI NAL units that contain a scalable nesting SEI message with `sn_ols_flag` equal to 1 and `sn_subpic_flag` equal to 0 that applies to the `targetOlsIdx`-th OLS (i.e., there is at least one value of `i` in the range of 0 to `sn_num_olss_minus1`, inclusive, such that `NestingOlsIdx[i]` is equal to `targetOlsIdx`), the following applies in the order listed:

- i. For each scalable-nested BP or DUI SEI message in such an SEI NAL unit `seiNalUnitA`, generate a non-scalable-nested SEI message with the same `payloadType` and SEI payload and include it in an SEI NAL unit in the PU containing `seiNalUnitA` in `outBitstream`.
- ii. When `general_same_pic_timing_in_all_ols_flag` is equal to 0, for each scalable-nested PT SEI message in such an SEI NAL unit `seiNalUnitA`, generate a non-scalable-nested SEI message with the same SEI payload and include it in an SEI NAL unit in the PU containing `seiNalUnitA` in `outBitstream`.
- iii. When multiple SEI messages contained in a particular such SEI NAL unit `seiNalUnitA` were made non-scalable-nested SEI messages, those non-scalable-nested SEI messages are included in one SEI NAL unit.
- iv. When there is no OLS, other than the target OLS in the OLSs to which the scalable-nested SEI messages in such an SEI NAL unit `seiNalUnitA` apply, that contains layers that are all included in the list `LayerIdInOls[targetOlsIdx]`, remove the SEI NAL unit `seiNalUnitA` from `outBitstream`.]]

FIG. 1 is a block diagram showing an example video processing system 1900 in which various techniques disclosed herein may be implemented. Various implementations may include some or all of the components of the system 1900. The system 1900 may include input 1902 for receiving video content. The video content may be received in a raw or uncompressed format, e.g., 8 or 10 bit multi-component pixel values, or may be in a compressed or encoded format. The input 1902 may represent a network interface, a peripheral bus interface, or a storage interface. Examples of network interface include wired interfaces such as Ethernet, passive optical network (PON), etc. and wireless interfaces such as Wi-Fi or cellular interfaces.

The system 1900 may include a coding component 1904 that may implement the various coding or encoding methods

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described in the present document. The coding component 1904 may reduce the average bitrate of video from the input 1902 to the output of the coding component 1904 to produce a coded representation of the video. The coding techniques are therefore sometimes called video compression or video transcoding techniques. The output of the coding component 1904 may be either stored, or transmitted via a communication connected, as represented by the component 1906. The stored or communicated bitstream (or coded) representation of the video received at the input 1902 may be used by the component 1908 for generating pixel values or displayable video that is sent to a display interface 1910. The process of generating user-viewable video from the bitstream representation is sometimes called video decompression. Furthermore, while certain video processing operations are referred to as "coding" operations or tools, it will be appreciated that the coding tools or operations are used at an encoder and corresponding decoding tools or operations that reverse the results of the coding will be performed by a decoder.

Examples of a peripheral bus interface or a display interface may include universal serial bus (USB) or high definition multimedia interface (HDMI) or Displayport, and so on. Examples of storage interfaces include serial advanced technology attachment (SATA), peripheral component interconnect (PCI), integrated drive electronics (IDE) interface, and the like. The techniques described in the present document may be embodied in various electronic devices such as mobile phones, laptops, smartphones or other devices that are capable of performing digital data processing and/or video display.

FIG. 2 is a block diagram of a video processing apparatus 3600. The apparatus 3600 may be used to implement one or more of the methods described herein. The apparatus 3600 may be embodied in a smartphone, tablet, computer, Internet of Things (IoT) receiver, and so on. The apparatus 3600 may include one or more processors 3602, one or more memories 3604 and video processing hardware 3606. The processor(s) 3602 may be configured to implement one or more methods described in the present document. The memory (memories) 3604 may be used for storing data and code used for implementing the methods and techniques described herein. The video processing hardware 3606 may be used to implement, in hardware circuitry, some techniques described in the present document.

FIG. 4 is a block diagram that illustrates an example video coding system 100 that may utilize the techniques of this disclosure.

As shown in FIG. 4, video coding system 100 may include a source device 110 and a destination device 120. Source device 110 generates encoded video data which may be referred to as a video encoding device. Destination device 120 may decode the encoded video data generated by source device 110 which may be referred to as a video decoding device.

Source device 110 may include a video source 112, a video encoder 114, and an input/output (I/O) interface 116.

Video source 112 may include a source such as a video capture device, an interface to receive video data from a video content provider, and/or a computer graphics system for generating video data, or a combination of such sources. The video data may comprise one or more pictures. Video encoder 114 encodes the video data from video source 112 to generate a bitstream. The bitstream may include a sequence of bits that form a coded representation of the video data. The bitstream may include coded pictures and associated data. The coded picture is a coded representation

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of a picture. The associated data may include sequence parameter sets, picture parameter sets, and other syntax structures. I/O interface **116** may include a modulator/demodulator (modem) and/or a transmitter. The encoded video data may be transmitted directly to destination device **120** via I/O interface **116** through network **130a**. The encoded video data may also be stored onto a storage medium/server **130b** for access by destination device **120**.

Destination device **120** may include an I/O interface **126**, a video decoder **124**, and a display device **122**.

I/O interface **126** may include a receiver and/or a modem. I/O interface **126** may acquire encoded video data from the source device **110** or the storage medium/ server **130b**. Video decoder **124** may decode the encoded video data. Display device **122** may display the decoded video data to a user. Display device **122** may be integrated with the destination device **120**, or may be external to destination device **120** which be configured to interface with an external display device.

Video encoder **114** and video decoder **124** may operate according to a video compression standard, such as the High Efficiency Video Coding (HEVC) standard, Versatile Video Coding (VVC) standard and other current and/or further standards.

FIG. **5** is a block diagram illustrating an example of video encoder **200**, which may be video encoder **114** in the system **100** illustrated in FIG. **4**.

Video encoder **200** may be configured to perform any or all of the techniques of this disclosure. In the example of FIG. **5**, video encoder **200** includes a plurality of functional components. The techniques described in this disclosure may be shared among the various components of video encoder **200**. In some examples, a processor may be configured to perform any or all of the techniques described in this disclosure.

The functional components of video encoder **200** may include a partition unit **201**, a prediction unit **202** which may include a mode select unit **203**, a motion estimation unit **204**, a motion compensation unit **205** and an intra prediction unit **206**, a residual generation unit **207**, a transform unit **208**, a quantization unit **209**, an inverse quantization unit **210**, an inverse transform unit **211**, a reconstruction unit **212**, a buffer **213**, and an entropy encoding unit **214**.

In other examples, video encoder **200** may include more, fewer, or different functional components. In an example, prediction unit **202** may include an intra block copy (IBC) unit. The IBC unit may perform prediction in an IBC mode in which at least one reference picture is a picture where the current video block is located.

Furthermore, some components, such as motion estimation unit **204** and motion compensation unit **205** may be highly integrated, but are represented in the example of FIG. **5** separately for purposes of explanation.

Partition unit **201** may partition a picture into one or more video blocks. Video encoder **200** and video decoder **300** may support various video block sizes.

Mode select unit **203** may select one of the coding modes, intra or inter, e.g., based on error results, and provide the resulting intra- or inter-coded block to a residual generation unit **207** to generate residual block data and to a reconstruction unit **212** to reconstruct the encoded block for use as a reference picture. In some example, mode select unit **203** may select a combination of intra and inter prediction (CIIP) mode in which the prediction is based on an inter prediction signal and an intra prediction signal. Mode select unit **203**

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may also select a resolution for a motion vector (e.g., a sub-pixel or integer pixel precision) for the block in the case of inter-prediction.

To perform inter prediction on a current video block, motion estimation unit **204** may generate motion information for the current video block by comparing one or more reference frames from buffer **213** to the current video block. Motion compensation unit **205** may determine a predicted video block for the current video block based on the motion information and decoded samples of pictures from buffer **213** other than the picture associated with the current video block.

Motion estimation unit **204** and motion compensation unit **205** may perform different operations for a current video block, for example, depending on whether the current video block is in an I slice, a P slice, or a B slice.

In some examples, motion estimation unit **204** may perform uni-directional prediction for the current video block, and motion estimation unit **204** may search reference pictures of list 0 or list 1 for a reference video block for the current video block. Motion estimation unit **204** may then generate a reference index that indicates the reference picture in list 0 or list 1 that contains the reference video block and a motion vector that indicates a spatial displacement between the current video block and the reference video block. Motion estimation unit **204** may output the reference index, a prediction direction indicator, and the motion vector as the motion information of the current video block. Motion compensation unit **205** may generate the predicted video block of the current block based on the reference video block indicated by the motion information of the current video block.

In other examples, motion estimation unit **204** may perform bi-directional prediction for the current video block, motion estimation unit **204** may search the reference pictures in list 0 for a reference video block for the current video block and may also search the reference pictures in list 1 for another reference video block for the current video block. Motion estimation unit **204** may then generate reference indexes that indicate the reference pictures in list 0 and list 1 containing the reference video blocks and motion vectors that indicate spatial displacements between the reference video blocks and the current video block. Motion estimation unit **204** may output the reference indexes and the motion vectors of the current video block as the motion information of the current video block. Motion compensation unit **205** may generate the predicted video block of the current video block based on the reference video blocks indicated by the motion information of the current video block.

In some examples, motion estimation unit **204** may output a full set of motion information for decoding processing of a decoder.

In some examples, motion estimation unit **204** may do not output a full set of motion information for the current video. Rather, motion estimation unit **204** may signal the motion information of the current video block with reference to the motion information of another video block. For example, motion estimation unit **204** may determine that the motion information of the current video block is sufficiently similar to the motion information of a neighboring video block.

In one example, motion estimation unit **204** may indicate, in a syntax structure associated with the current video block, a value that indicates to the video decoder **300** that the current video block has the same motion information as the another video block.

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In another example, motion estimation unit **204** may identify, in a syntax structure associated with the current video block, another video block and a motion vector difference (MVD). The motion vector difference indicates a difference between the motion vector of the current video block and the motion vector of the indicated video block. The video decoder **300** may use the motion vector of the indicated video block and the motion vector difference to determine the motion vector of the current video block.

As discussed above, video encoder **200** may predictively signal the motion vector. Two examples of predictive signaling techniques that may be implemented by video encoder **200** include advanced motion vector prediction (AMVP) and merge mode signaling.

Intra prediction unit **206** may perform intra prediction on the current video block. When intra prediction unit **206** performs intra prediction on the current video block, intra prediction unit **206** may generate prediction data for the current video block based on decoded samples of other video blocks in the same picture. The prediction data for the current video block may include a predicted video block and various syntax elements.

Residual generation unit **207** may generate residual data for the current video block by subtracting (e.g., indicated by the minus sign) the predicted video block(s) of the current video block from the current video block. The residual data of the current video block may include residual video blocks that correspond to different sample components of the samples in the current video block.

In other examples, there may be no residual data for the current video block for the current video block, for example in a skip mode, and residual generation unit **207** may not perform the subtracting operation.

Transform processing unit **208** may generate one or more transform coefficient video blocks for the current video block by applying one or more transforms to a residual video block associated with the current video block.

After transform processing unit **208** generates a transform coefficient video block associated with the current video block, quantization unit **209** may quantize the transform coefficient video block associated with the current video block based on one or more quantization parameter (QP) values associated with the current video block.

Inverse quantization unit **210** and inverse transform unit **211** may apply inverse quantization and inverse transforms to the transform coefficient video block, respectively, to reconstruct a residual video block from the transform coefficient video block. Reconstruction unit **212** may add the reconstructed residual video block to corresponding samples from one or more predicted video blocks generated by the prediction unit **202** to produce a reconstructed video block associated with the current block for storage in the buffer **213**.

After reconstruction unit **212** reconstructs the video block, loop filtering operation may be performed reduce video blocking artifacts in the video block.

Entropy encoding unit **214** may receive data from other functional components of the video encoder **200**. When entropy encoding unit **214** receives the data, entropy encoding unit **214** may perform one or more entropy encoding operations to generate entropy encoded data and output a bitstream that includes the entropy encoded data.

FIG. 6 is a block diagram illustrating an example of video decoder **300** which may be video decoder **124** in the system **100** illustrated in FIG. 4.

The video decoder **300** may be configured to perform any or all of the techniques of this disclosure. In the example of

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FIG. 5, the video decoder **300** includes a plurality of functional components. The techniques described in this disclosure may be shared among the various components of the video decoder **300**. In some examples, a processor may be configured to perform any or all of the techniques described in this disclosure.

In the example of FIG. 6, video decoder **300** includes an entropy decoding unit **301**, a motion compensation unit **302**, an intra prediction unit **303**, an inverse quantization unit **304**, an inverse transformation unit **305**, and a reconstruction unit **306** and a buffer **307**. Video decoder **300** may, in some examples, perform a decoding pass generally reciprocal to the encoding pass described with respect to video encoder **200** (FIG. 5).

Entropy decoding unit **301** may retrieve an encoded bitstream. The encoded bitstream may include entropy coded video data (e.g., encoded blocks of video data). Entropy decoding unit **301** may decode the entropy coded video data, and from the entropy decoded video data, motion compensation unit **302** may determine motion information including motion vectors, motion vector precision, reference picture list indexes, and other motion information. Motion compensation unit **302** may, for example, determine such information by performing the AMVP and merge mode.

Motion compensation unit **302** may produce motion compensated blocks, possibly performing interpolation based on interpolation filters. Identifiers for interpolation filters to be used with sub-pixel precision may be included in the syntax elements.

Motion compensation unit **302** may use interpolation filters as used by video encoder **200** during encoding of the video block to calculate interpolated values for sub-integer pixels of a reference block. Motion compensation unit **302** may determine the interpolation filters used by video encoder **200** according to received syntax information and use the interpolation filters to produce predictive blocks.

Motion compensation unit **302** may use some of the syntax information to determine sizes of blocks used to encode frame(s) and/or slice(s) of the encoded video sequence, partition information that describes how each macroblock of a picture of the encoded video sequence is partitioned, modes indicating how each partition is encoded, one or more reference frames (and reference frame lists) for each inter-encoded block, and other information to decode the encoded video sequence.

Intra prediction unit **303** may use intra prediction modes for example received in the bitstream to form a prediction block from spatially adjacent blocks. Inverse quantization unit **304** inverse quantizes, i.e., de-quantizes, the quantized video block coefficients provided in the bitstream and decoded by entropy decoding unit **301**. Inverse transform unit **305** applies an inverse transform.

Reconstruction unit **306** may sum the residual blocks with the corresponding prediction blocks generated by motion compensation unit **302** or intra-prediction unit **303** to form decoded blocks. If desired, a deblocking filter may also be applied to filter the decoded blocks in order to remove blockiness artifacts. The decoded video blocks are then stored in buffer **307**, which provides reference blocks for subsequent motion compensation/intra prediction and also produces decoded video for presentation on a display device.

A listing of solutions preferred by some embodiments is provided next.

A first set of solutions is provided next. The following solutions show example embodiments of techniques discussed in the previous section (e.g., items 1-9).

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1. A method of video processing (e.g., method **600** in FIG. 3), comprising performing (**602**) a conversion between a video comprising one or more video layers comprising one or more video pictures and a coded representation of the video, wherein the coded representation conforms to a format rule related to extraction of a sub-bitstream from the coded representation. 5
2. The method of solution 1, further comprising: extracting the sub-bitstream from the coded representation according to the format rule. 10

The following solutions show example embodiments of techniques discussed in the previous section (e.g., item 1)

3. The method of any of solutions 1-2, wherein, during extracting the sub-bitstream, a target id used for the extracting is allowed to be between range 0 to a value of a syntax field indicating in a video parameter set for the coded representation. 15

The following solutions show example embodiments of techniques discussed in the previous section (e.g., item 2)

4. The method of any of solutions 1-3, wherein the sub-bitstream is extracted without removing an access unit delimiter network abstraction layer (AUD NAL) from an output bitstream according to a layer id. 20

The following solutions show example embodiments of techniques discussed in the previous section (e.g., item 3)

5. The method of any of solutions 1-4, wherein the sub-bitstream is extracted by selectively removing network abstraction layer units that include a scalable nesting supplementary enhancement information message that are not applicable to output layers being extracted. 25

The following solutions show example embodiments of techniques discussed in the previous section (e.g., item 5)

6. The method of any of solutions 1-5, wherein the sub-bitstream is extracted by constraining the extracting to generate a non-scalable nested supplemental enhancement information (SEI) from a scalable nesting SEI message using a flag for output layer set being set and a flag for subpicture being disabled. 30

The following solutions show example embodiments of techniques discussed in the previous section (e.g., item 6)

7. The method of any of solutions 1-6, wherein the sub-bitstream is extracted according to a rule specifying extraction of multiple scalable-nested supplemental enhancement information (SEI) messages from a single SEI network abstraction layer unit. 35

The following solutions show example embodiments of techniques discussed in the previous section (e.g., item 7)

8. The method of any of solutions 1-7, wherein the sub-bitstream is extracted according to a rule that removes, from the coded representation, all supplemental enhancement information network abstraction layer SEI NAL units from which some SEI messages have been extracted and included as non-scalable-nested SEI messages. 40

The following solutions show example embodiments of techniques discussed in the previous section (e.g., item 8-13).

10. A video processing method comprising: performing a conversion between a video and a coded representation of the video, wherein the coded representation comprises one or more sub-bitstreams; wherein the conversion is performed according to a rule that specifies a relationship between a sub-bitstream extraction process and one or more syntax elements of the coded representation. 45

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11. The method of solution 10, wherein the rule specifies to remove all supplemental enhancement information (SEI) network abstraction layer units (NAL) that contain an SEI message that apply to a picture or a subpicture for which the video coded layer is removed in case that one or more two conditions are true: (1) a type of the NAL unit is a specific type or (2) temporal identifier of the video coded layer meets a specific condition.
 12. The method of any of solutions 10-11, wherein the rule specifies that, in case that layer identifiers in target output layer do not include all values of layer identifiers in the coded representation, the process additional removes all SEI NAL units that contain a non-scalable-nested SEI message with payloadType equal to 203.
 13. The method of any of solutions 1-12, wherein the performing the conversion comprises encoding the video into the coded representation.
 14. The method of any of solutions 1-12, wherein the performing the conversion comprises parsing and decoding the coded representation to generate the video.
 15. A video decoding apparatus comprising a processor configured to implement a method recited in one or more of solutions 1 to 14.
 16. A video encoding apparatus comprising a processor configured to implement a method recited in one or more of solutions 1 to 14.
 17. A computer program product having computer code stored thereon, the code, when executed by a processor, causes the processor to implement a method recited in any of solutions 1 to 14.
 18. A method, apparatus or system described in the present document.
- A second set of solutions show example embodiments of techniques discussed in the previous section (e.g., items 10 and 11).
1. A method of processing video data (e.g., method **700** as shown in FIG. 7A), comprising: performing (**702**) a conversion between a video and a bitstream of the video, wherein the bitstream includes network abstraction layer (NAL) units for multiple video layers according to a rule; wherein the rule defines a sub-bitstream extraction process by which NAL units are removed from the bitstream to generate an output bitstream, and wherein the rule specifies to remove all supplemental enhancement information (SEI) NAL units that contain a non-scalable-nested SEI message with a particular payload type, responsive to a list of NAL unit header layer identifier values in an output layer set (OLS) with a target OLS index not including all values of NAL unit header layer identifiers in all video coding layer (VCL) NAL units in the bitstream that is input to the sub-bitstream extraction process.
 2. The method of solution 1, wherein the non-scalable-nested SEI message with the particular payload type corresponds to a subpicture level information SEI message.
 3. The method of solution 1, wherein the particular payload type is equal to 203.
 4. The method of solution 1, wherein the non-scalable-nested SEI message with the particular payload type corresponds to a decoding unit information message.
 5. The method of solution 1, wherein the particular payload type is equal to 130.
 6. A method of processing video data (e.g., method **710** as shown in FIG. 7B), comprising: performing a conver-

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- sion between a video and a bitstream of the video, wherein the bitstream is separable into one or more sub-bitstreams according to a rule that specifies a sub-bitstream extraction process to generate an output bitstream, and wherein the rule specifies whether or how to remove, based on a type of a video coding layer (VCL) network abstraction layer (NAL) unit and a temporal identifier of a video coding layer associated with the VCL NAL unit, all supplemental enhancement information (SEI) network abstraction layer (NAL) units that contain an SEI message that apply to a picture or a subpicture for which the VCL NAL unit is removed during the sub-bitstream extraction process.
7. The method of solution 6, wherein the rule specifies to remove all SEI NAL units responsive to (1) the type of the VCL NAL unit equal to TRAIL_NUT, STSA_NUT, RADL_NUT, or RASL_NUT, or GDR_NUT and (2) the temporal identifier of the video coding layer is greater than or equal to a number of sublayers in the video coding layer in an output layer set.
 8. The method of solution 7, wherein, for the type of the VCL NAL unit equal to GDR_NUT, a syntax field indicative of a recovery point of decoded pictures in output order has a value greater than 0.
 9. The method of solution 6, wherein the rule specifies to keep a SEI NAL unit in the output bitstream.
 10. The method of any of solutions 1 to 9, wherein the conversion includes encoding the video into the bitstream.
 11. The method of any of solutions 1 to 9, wherein the conversion includes decoding the video from the bitstream.
 12. The method of any of solutions 1 to 9, wherein the conversion includes generating the bitstream from the video, and the method further comprises: storing the bitstream in a non-transitory computer-readable recording medium.
 13. A video processing apparatus comprising a processor configured to implement a method recited in any one or more of solutions 1 to 12.
 14. A method of storing a bitstream of a video, comprising, a method recited in any one of solutions 1 to 12, and further including storing the bitstream to a non-transitory computer-readable recording medium.
 15. A computer readable medium storing program code that, when executed, causes a processor to implement a method recited in any one or more of solutions 1 to 12.
 16. A computer readable medium that stores a coded representation or a bitstream generated according to any of the above described methods.
 17. A video processing apparatus for storing a bitstream, wherein the video processing apparatus is configured to implement a method recited in any one or more of solutions 1 to 12.
- A third set of solutions show example embodiments of techniques discussed in the previous section (e.g., items 12 and 13).
1. A method of processing video data (e.g., method 800 as shown in FIG. 8), comprising: performing 802 a conversion between a video and a bitstream of the video according to a rule, wherein the bitstream includes network abstraction layer (NAL) units for multiple video layers according to a rule; wherein the rule defines that a sub-bitstream extraction process to generate an output bitstream comprising an output layer set (OLS) includes one or more operations that are selectively performed responsive to the following condi-

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- tions: (1) a list of NAL unit header layer identifier values in the OLS does not includes all values of NAL unit header layer identifiers in all video coding layer (VCL) NAL units in the bitstream, and (2) the output bitstream containing a supplemental enhancement information (SEI) NAL unit that contains a scalable-nesting SEI message.
2. The method of solution 1, wherein the scalable-nesting SEI message is associated with a first flag equal to a particular value that applies to specific OLSs or specific layers that have a same set of layers as in the output bitstream.
 3. The method of solution 2, wherein the first flag equal to 0 specifies that scalable-nested SEI messages that apply to the specific OLSs or the specific layers apply to all subpictures of the specified OLSs or the specific layers.
 4. The method of solution 1, wherein the scalable-nesting SEI message is associated with a second flag indicative of whether the scalable-nesting SEI message applies to specific OLSs or specific layers.
 5. The method of any of solutions 1-4, wherein the one or more operations include generating an additional SEI NAL unit.
 6. The method of solution 5, wherein the one or more operations include including the additional SEI NAL unit in a picture unit containing the SEI NAL unit.
 7. The method of solution 6, wherein the additional SEI NAL unit is immediately after the SEI NAL unit.
 8. The method of any of solution 6, wherein the one or more operations further include extracting scalable-nested SEI messages from the scalable-nesting SEI message and including the scalable-nested SEI messages in the additional SEI NAL unit.
 9. The method of solution 1, wherein the one or more operations include removing the SEI NAL unit from the output bitstream.
 10. The method of solution 1, wherein the one or more operations include keeping the SEI NAL unit in the output bitstream.
 11. The method of any of solutions 1 to 10, wherein the conversion includes encoding the video into the bitstream.
 12. The method of any of solutions 1 to 10, wherein the conversion includes decoding the video from the bitstream.
 13. The method of any of solutions 1 to 10, wherein the conversion includes generating the bitstream from the video, and the method further comprises: storing the bitstream in a non-transitory computer-readable recording medium.
 14. A video processing apparatus comprising a processor configured to implement a method recited in any one or more of solutions 1 to 13.
 15. A method of storing a bitstream of a video, comprising, a method recited in any one of solutions 1 to 13, and further including storing the bitstream to a non-transitory computer-readable recording medium.
 16. A computer readable medium storing program code that, when executed, causes a processor to implement a method recited in any one or more of solutions 1 to 13.
 17. A computer readable medium that stores a bitstream generated according to any of the above described methods.

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18. A video processing apparatus for storing a bitstream, wherein the video processing apparatus is configured to implement a method recited in any one or more of solutions 1 to 13.

The disclosed and other solutions, examples, embodiments, modules and the functional operations described in this document can be implemented in digital electronic circuitry, or in computer software, firmware, or hardware, including the structures disclosed in this document and their structural equivalents, or in combinations of one or more of them. The disclosed and other embodiments can be implemented as one or more computer program products, i.e., one or more modules of computer program instructions encoded on a computer readable medium for execution by, or to control the operation of, data processing apparatus. The computer readable medium can be a machine-readable storage device, a machine-readable storage substrate, a memory device, a composition of matter effecting a machine-readable propagated signal, or a combination of one or more of them. The term "data processing apparatus" encompasses all apparatus, devices, and machines for processing data, including by way of example a programmable processor, a computer, or multiple processors or computers. The apparatus can include, in addition to hardware, code that creates an execution environment for the computer program in question, e.g., code that constitutes processor firmware, a protocol stack, a database management system, an operating system, or a combination of one or more of them. A propagated signal is an artificially generated signal, e.g., a machine-generated electrical, optical, or electromagnetic signal, that is generated to encode information for transmission to suitable receiver apparatus.

A computer program (also known as a program, software, software application, script, or code) can be written in any form of programming language, including compiled or interpreted languages, and it can be deployed in any form, including as a stand-alone program or as a module, component, subroutine, or other unit suitable for use in a computing environment. A computer program does not necessarily correspond to a file in a file system. A program can be stored in a portion of a file that holds other programs or data (e.g., one or more scripts stored in a markup language document), in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, sub programs, or portions of code). A computer program can be deployed to be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network.

The processes and logic flows described in this document can be performed by one or more programmable processors executing one or more computer programs to perform functions by operating on input data and generating output. The processes and logic flows can also be performed by, and apparatus can also be implemented as, special purpose logic circuitry, e.g., an field programmable gate array (FPGA) or an application specific integrated circuit (ASIC).

Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor will receive instructions and data from a read only memory or a random-access memory or both. The essential elements of a computer are a processor for performing instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive data from or transfer data to, or

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both, one or more mass storage devices for storing data, e.g., magnetic, magneto optical disks, or optical disks. However, a computer need not have such devices. Computer readable media suitable for storing computer program instructions and data include all forms of non-volatile memory, media and memory devices, including by way of example semiconductor memory devices, e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto optical disks; and compact disc, read-only memory (CD ROM) and digital versatile disc read-only memory (DVD-ROM) disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.

While this patent document contains many specifics, these should not be construed as limitations on the scope of any subject matter or of what may be claimed, but rather as descriptions of features that may be specific to particular embodiments of particular techniques. Certain features that are described in this patent document in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. Moreover, the separation of various system components in the embodiments described in this patent document should not be understood as requiring such separation in all embodiments.

Only a few implementations and examples are described and other implementations, enhancements and variations can be made based on what is described and illustrated in this patent document.

What is claimed is:

1. A method of processing video data, comprising: performing a conversion between a video including one or more layers and a bitstream of the video according to a rule, wherein the rule specifies a sub-bitstream extraction process by which network abstraction layer (NAL) units are removed from an input bitstream by the sub-bitstream extraction process to generate an output sub-bitstream, wherein the rule further specifies that when a first set of conditions are satisfied, all supplemental enhancement information (SEI) NAL units that contain a non-scalable-nested SEI message with particular payload types are removed, wherein the first set of conditions comprises that a list of NAL unit header layer identifier values in an output layer set (OLS) with a target OLS index does not include all values of NAL unit header layer identifiers in all video coding layer (VCL) NAL units in the input bitstream, and wherein the non-scalable-nested SEI message with the particular payload types comprises a non-scalable-

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nested SEI message with a first particular payload type corresponding to a subpicture level information SEI message,

wherein the rule further specifies that when a second set of conditions are satisfied, a new SEI NAL unit is generated, and scalable-nested SEI messages are extracted from the particular scalable nesting SEI message and included in the new SEI NAL unit,

wherein the second set of conditions comprises the first set of conditions and that the output sub-bitstream contains a first supplemental enhancement information (SEI) NAL unit that contains a particular scalable nesting SEI message.

2. The method of claim 1, wherein the particular scalable nesting SEI message is a scalable nesting SEI message with a first flag equal to a first value and a second flag equal to a second value that applies to the OLS with the target OLS index, wherein the first flag equal to the first value indicates that the scalable nesting SEI message applies to specific output layer sets (OLSs), and wherein the second flag equal to the second value indicates that the scalable nesting SEI message applies to all subpictures.

3. The method of claim 1, wherein the scalable-nested SEI messages are included in the new SEI NAL unit as non-scalable-nested SEI messages, wherein the new SEI NAL unit is included in a picture unit (PU) containing the first SEI NAL unit, wherein the new SEI NAL unit is included immediately after the first SEI NAL unit, and wherein the first SEI NAL unit is removed from the output sub-bitstream after the scalable-nested SEI messages being included in the new SEI NAL unit.

4. The method of claim 1, wherein the first particular payload type is equal to 203.

5. The method of claim 1, wherein the non-scalable-nested SEI message with the particular payload types further comprises a non-scalable-nested SEI message with a second particular payload type corresponding to a decoding unit information message, and wherein the second particular payload type is equal to 130.

6. The method of claim 1, wherein the rule further specifies that NAL units satisfying a first set of conditions are removed, wherein the first set of conditions comprises that: 1) NAL units have nuh_layer_id not included in the list of NAL unit header layer identifier values in the OLS with the target OLS index, 2) NAL units are not decoding capability information (DCI) NAL units, video parameter set (VPS) NAL units, access unit delimiter (AUD) NAL units, or end of bitstream (EOB) NAL units, and 3) NAL units are not NAL units containing a particular non-scalable-nested SEI message.

7. The method of claim 1, wherein the rule further specifies that all VCL NAL units satisfying a second set of conditions are removed, wherein the second set of conditions comprises that nal_unit_type is equal to TRAIL_NUT, STSA_NUT, RADL_NUT, or RASL_NUT, or nal_unit_type is equal to GDR_NUT and associated ph_recovery_poc_cnt is greater than 0.

8. The method of claim 1, wherein the conversion includes encoding the video into the bitstream.

9. The method of claim 1, wherein the conversion includes decoding the video from the bitstream.

10. An apparatus for processing video data comprising a processor and a non-transitory memory with instructions thereon, wherein the instructions upon execution by the processor, cause the processor to:

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perform a conversion between a video including one or more layers and a bitstream of the video according to a rule,

wherein the rule specifies a sub-bitstream extraction process by which network abstraction layer (NAL) units are removed from an input bitstream by the sub-bitstream extraction process to generate an output sub-bitstream,

wherein the rule further specifies that when a first set of conditions are satisfied, all supplemental enhancement information (SEI) NAL units that contain a non-scalable-nested SEI message with particular payload types are removed,

wherein the first set of conditions comprises that a list of NAL unit header layer identifier values in an output layer set (OLS) with a target OLS index does not include all values of NAL unit header layer identifiers in all video coding layer (VCL) NAL units in the input bitstream, and

wherein the non-scalable-nested SEI message with the particular payload types comprises a non-scalable-nested SEI message with a first particular payload type corresponding to a subpicture level information SEI message,

wherein the rule further specifies that when a second set of conditions are satisfied, a new SEI NAL unit is generated, and scalable-nested SEI messages are extracted from the particular scalable nesting SEI message and included in the new SEI NAL unit,

wherein the second set of conditions comprises the first set of conditions and that the output sub-bitstream contains a first supplemental enhancement information (SEI) NAL unit that contains a particular scalable nesting SEI message.

11. The apparatus of claim 10, wherein the particular scalable nesting SEI message is a scalable nesting SEI message with a first flag equal to a first value and a second flag equal to a second value that applies to the OLS with the target OLS index, wherein the first flag equal to the first value indicates that the scalable nesting SEI message applies to specific output layer sets (OLSs), wherein the second flag equal to the second value indicates that the scalable nesting SEI message applies to all subpictures, wherein the scalable-nested SEI messages are included in the new SEI NAL unit as non-scalable-nested SEI messages, wherein the new SEI NAL unit is included in a picture unit (PU) containing the first SEI NAL unit, wherein the new SEI NAL unit is included immediately after the first SEI NAL unit, and wherein the first SEI NAL unit is removed from the output sub-bitstream after the scalable-nested SEI messages being included in the new SEI NAL unit.

12. The apparatus of claim 10, wherein the rule further specifies that NAL units satisfying a first set of conditions are removed, wherein the first set of conditions comprises that: 1) NAL units have nuh_layer_id not included in the list of NAL unit header layer identifier values in the OLS with the target OLS index, 2) NAL units are not decoding capability information (DCI) NAL units, video parameter set (VPS) NAL units, access unit delimiter (AUD) NAL units, or end of bitstream (EOB) NAL units, and 3) NAL units are not NAL units containing a particular non-scalable-nested SEI message, and wherein the rule further specifies that all VCL NAL units satisfying a second set of conditions are removed, wherein the second set of conditions comprises that nal_unit_type is equal to TRAIL_NUT, STSA_NUT,

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RADL_NUT, or RASL_NUT, or nal_unit_type is equal to GDR_NUT and associated ph_recovery_poc_cnt is greater than 0.

13. A non-transitory computer-readable storage medium storing instructions that cause a processor to:

perform a conversion between a video including one or more layers and a bitstream of the video according to a rule, wherein the rule specifies a sub-bitstream extraction process by which network abstraction layer (NAL) units are removed from an input bitstream by the sub-bitstream extraction process to generate an output sub-bitstream,

wherein the rule further specifies that when a first set of conditions are satisfied, all supplemental enhancement information (SEI) NAL units that contain a non-scalable-nested SEI message with particular payload types are removed,

wherein the first set of conditions comprises that a list of NAL unit header layer identifier values in an output layer set (OLS) with a target OLS index does not include all values of NAL unit header layer identifiers in all video coding layer (VCL) NAL units in the input bitstream, and

wherein the non-scalable-nested SEI message with the particular payload types comprises a non-scalable-nested SEI message with a first particular payload type corresponding to a subpicture level information SEI message,

wherein the rule further specifies that when a second set of conditions are satisfied, a new SEI NAL unit is generated, and scalable-nested SEI messages are extracted from the particular scalable nesting SEI message and included in the new SEI NAL unit,

wherein the second set of conditions comprises the first set of conditions and that the output sub-bitstream contains a first supplemental enhancement information (SEI) NAL unit that contains a particular scalable nesting SEI message.

14. The non-transitory computer-readable storage medium of claim 13, wherein the particular scalable nesting SEI message is a scalable nesting SEI message with a first flag equal to a first value and a second flag equal to a second value that applies to the OLS with the target OLS index,

wherein the first flag equal to the first value indicates that the scalable nesting SEI message applies to specific output layer sets (OLSs),

wherein the second flag equal to the second value indicates that the scalable nesting SEI message applies to all subpictures,

wherein the scalable-nested SEI messages are included in the new SEI NAL unit as non-scalable-nested SEI messages,

wherein the new SEI NAL unit is included in a picture unit (PU) containing the first SEI NAL unit, wherein the new SEI NAL unit is included immediately after the first SEI NAL unit, and

wherein the first SEI NAL unit is removed from the output sub-bitstream after the scalable-nested SEI messages being included in the new SEI NAL unit.

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15. The non-transitory computer-readable storage medium of claim 13, wherein the rule further specifies that NAL units satisfying a first set of conditions are removed, wherein the first set of conditions comprises that: 1) NAL units have nuh_layer_id not included in the list of NAL unit header layer identifier values in the OLS with the target OLS index, 2) NAL units are not decoding capability information (DCI) NAL units, video parameter set (VPS) NAL units, access unit delimiter (AUD) NAL units, or end of bitstream (EOB) NAL units, and 3) NAL units are not NAL units containing a particular non-scalable-nested SEI message, and wherein the rule further specifies that all VCL NAL units satisfying a second set of conditions are removed, wherein the second set of conditions comprises that nal_unit_type is equal to TRAIL_NUT, STSA_NUT, RADL_NUT, or RASL_NUT, or nal_unit_type is equal to GDR_NUT and associated ph_recovery_poc_cnt is greater than 0.

16. A method for storing a bitstream of a video, comprising:

generating the bitstream of the video including one or more layers according to a rule, and storing the bitstream in a non-transitory computer-readable recording medium,

wherein the rule specifies a sub-bitstream extraction process by which network abstraction layer (NAL) units are removed from an input bitstream by the sub-bitstream extraction process to generate an output sub-bitstream,

wherein the rule further specifies that when a first set of conditions are satisfied, all supplemental enhancement information (SEI) NAL units that contain a non-scalable-nested SEI message with particular payload types are removed,

wherein the first set of conditions comprises that a list of NAL unit header layer identifier values in an output layer set (OLS) with a target OLS index does not include all values of NAL unit header layer identifiers in all video coding layer (VCL) NAL units in the input bitstream, and

wherein the non-scalable-nested SEI message with the particular payload types comprises a non-scalable-nested SEI message with a first particular payload type corresponding to a subpicture level information SEI message,

wherein the rule further specifies that when a second set of conditions are satisfied, a new SEI NAL unit is generated, and scalable-nested SEI messages are extracted from the particular scalable nesting SEI message and included in the new SEI NAL unit,

wherein the second set of conditions comprises the first set of conditions and that the output sub-bitstream contains a first supplemental enhancement information (SEI) NAL unit that contains a particular scalable nesting SEI message.

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